



Climate justice through explainable graph neural networks: A spatiotemporal attention-based urban heat risk assessment under IPCC AR6 framework[☆]

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ABSTRACT

Climate change has intensified urban heat islands, disproportionately harming vulnerable populations, but existing heat risk assessments have limited policy use due to subjective weighting, spatial loss, and weak health validation. To overcome these limitations, this study developed an explainable spatiotemporal graph attention network (GAT) at a resolution of 500 m that incorporated climate-social variables from the IPCC AR6 framework to assess the heat risk in Seoul. Model reliability was tested using spatiotemporal cross-validation and distance correlation analysis with multi-agency health data, while interpretation consistency was cross-validated using an explainable artificial intelligence (XAI). The model demonstrated high predictive performance (spatiotemporal cross-validation: $R^2 = 0.968$) and empirical validity, achieving a distance correlation of 0.449 ($p < 0.001$) with heat-related illness rates, substantially exceeding conventional heat vulnerability indices. Spatial analysis revealed significant risk, with central Seoul having a 9.1% higher risk than southwest Seoul. The strong spatial concentration associated with climate hazards was transformed into a compound risk distribution via nonlinear socio-economic interactions. Multi-XAI analysis revealed a scale-dependent shift in the dominant risk factors, with urban exposure primarily responsible for between-district differences at the metropolitan scale and climate hazards dominating at the regional scale. Collectively, these results indicate that effective climate justice policies must adopt scale-differentiated strategies.

1. Introduction

The intensification of the urban heat island (UHI) effect due to climate change is an important component of climate justice. Heat waves exert disproportionate health impacts on the population depending on its socioeconomic characteristics, including age, income, and housing quality (IPCC, 2022). This unequal distribution of climate risks is exacerbated in megacities with populations exceeding 10 million due to the complex interaction between the physical urban structure and socioeconomic stratification (Sining and Jun, 2021). Therefore, equity-based adaptation strategies are required to protect vulnerable groups (Rosenzweig et al., 2018). To achieve this, heat vulnerability indices (HVIs) are widely used to assess urban heat vulnerability, but they often face methodological

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limitations. According to systematic literature reviews (Cheng et al., 2021; Niu et al., 2021), HVI studies are difficult to compare due to their subjectivity and lack of consistency in indicator selection and weighting methods. They are also vulnerable to the modifiable areal unit problem (MAUP), which causes statistical distortions based on spatial unit settings (Cheng et al., 2021). Niu et al. (2021) also found that only 28% of HVI studies performed validation using external health data and, among these validated studies, correlations were typically weak; for example, a Dallas study reported a R^2 of only 0.03 between vulnerability indices and mortality rates.

To overcome these limitations, artificial intelligence (AI) models have been proposed, yet they also have some issues that need to be resolved. While traditional linear models fail to capture nonlinear threshold effects between the temperature and population, deep learning approaches often suffer from the black-box problem, in which they lack the transparency required for policy implementation. In addition, although previous studies have integrated climate hazards into deep learning models (Li et al., 2024; Dong et al., 2014), few attempts have been made to explicitly model nonlinear interactions within the IPCC Sixth Assessment Report (AR6) framework. Furthermore, while explainable AI (XAI) can be used to interpret the results of deep-learning-based models (Lundberg and Lee, 2017; Ying et al., 2019), systematic cross-validation of consistency across different XAI methods remains limited in the spatiotemporal domain (Yuan et al., 2022).

To address these methodological limitations, this study develops an explainable spatiotemporal graph attention network (GAT) model at a resolution of 500 m that represents the first application of graph attention networks specifically structured around the IPCC AR6 risk assessment framework for urban heat. Rather than incorporating climate and socioeconomic variables as an undifferentiated input vector (Li et al., 2024; Dong et al., 2014), each of the four AR6 risk components is embedded as a dedicated graph encoder, connected through a cross-component attention mechanism that learns their multiplicative nonlinear interactions from data. This approach models how climate risks are amplified through compound effects on vulnerable populations, capturing a core principle of climate justice theory that posits that health impacts increase in a nonlinear fashion beyond threshold points. To ensure interpretability, we construct a systematic multi-XAI framework based on the use of Shapley additive explanations (SHAP), GNNExplainer, and integrated gradients. Finally, we verify the empirical validity of the model using distance correlation analysis with multi-agency health data, directly addressing the lack of health validation that has been highlighted in existing literature (Niu et al., 2021).

2. Materials and methods

2.1. Study area

Seoul Metropolitan City (37.42°N–37.70°N, 126.77°E–127.19°E), the capital of South Korea, was set as the study area for the present study. It was spatially discretized into 2634 analysis grids at a resolution of 500 m across 25 administrative districts (Fig. 1). As a representative East Asian megacity with approximately 9.7 million residents (average for 2015–2019), Seoul has a basin topography that is centered on the Han River and surrounded by mountains, thus creating natural conditions for the trapping of heat. This is exacerbated by rapid urbanization, which has increased the impervious surface coverage and anthropogenic heat emissions, thus intensifying heat accumulation (Hong et al., 2019). In addition, due to its rapid, compressed growth, Seoul has a complex urban

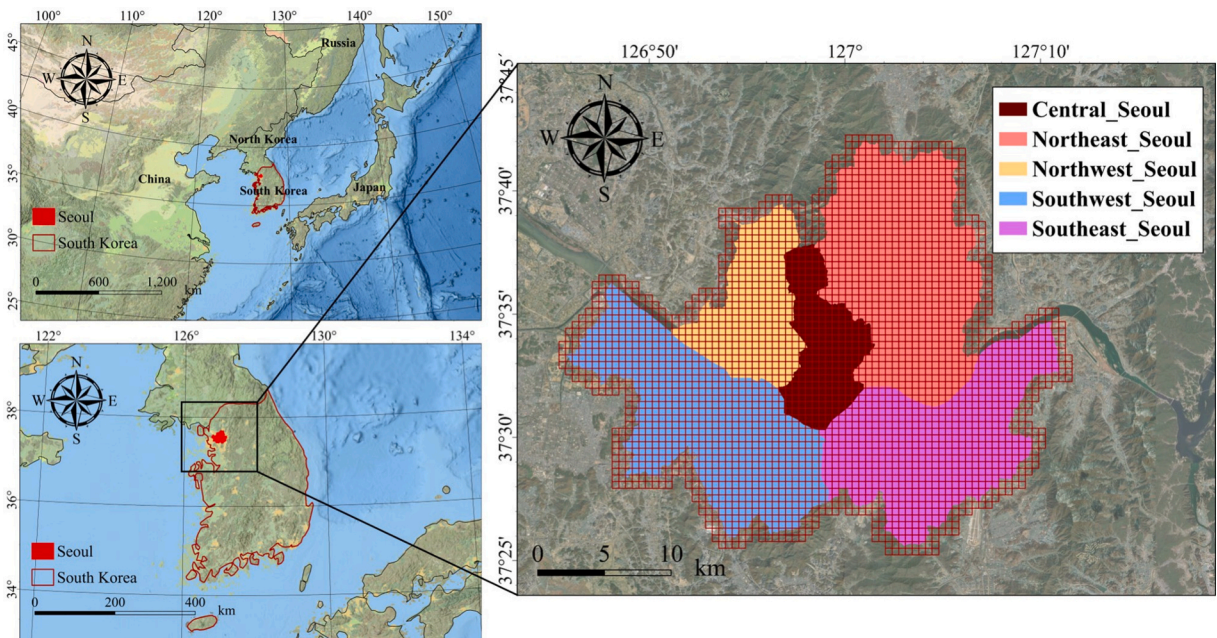


Fig. 1. Study area for the present study: Seoul, Korea. Seoul is divided into 2634 grids (resolution: 500 m) across 25 administrative districts. Five functional regions (central, northwest, northeast, southeast, southwest Seoul) are examined for regional-scale analysis.

structure. The Local Climate Zone (LCZ) classification provides a standardized framework for characterizing urban form by surface structure, land cover, and thermal properties (Stewart and Oke, 2012). Based on this classification, Yi et al. (2022) found that heat stress is most pronounced in LCZs 1 and 2 (compact high-rise and compact midrise zones, respectively), where the high density of buildings and low vegetation cover impede wind corridors and concentrate thermal stress. Ngarambe et al. (2020) reported that these UHI effects are amplified during heat waves, resulting in temperature differentials of up to 4.5 °C.

The socioeconomic spatial differentiation centered on the Han River is particularly suitable for the assessment of inequality in climate risk exposure. The Gangnam Belt (Gangnam, Seocho, and Songpa districts) forms a homogeneous high-socioeconomic cluster that is socially and economically distinct from the remaining 22 districts, leading to clear spatially distributed socioeconomic inequality (Han, 2022; Bae and Joo, 2020). In vulnerable areas outside this cluster, low housing quality, economic instability, and social isolation interact to aggravate exposure risks to extreme temperatures (Kang and Lyles, 2025). Consequently, the convergence of physical UHI effects and entrenched socioeconomic stratification means that Seoul is an optimal testbed for validating the proposed climate justice-based risk assessment framework.

On the microscale, our spatial framework employed 500 m × 500 m grids in accordance with the minimum diameter criterion for the LCZs proposed by Stewart and Oke (2012), thus effectively capturing urban microclimate heterogeneity while mitigating the MAUP (Reid et al., 2009; Fotheringham and Wong, 1991). For macroscopic regional analysis, we aggregated these grids into five functional regions (central, northwest, northeast, southeast, southwest) to reflect Seoul's spatial structure and land-use patterns (Ou et al., 2023; Jin and Kim, 2018; Kim and Baik, 2005) (Fig. 1). Additionally, for metropolitan-scale analysis, the grid-level results were aggregated into 25 administrative districts and subsequently compared with district-level health data to identify citywide risk disparities.

2.2. Data sources and variables

This study adopts the risk assessment framework of the IPCC AR6 (IPCC, 2022) for the architecture of the proposed GAT model. Within this framework, climate risk is defined as the result of dynamic interactions among hazards (potential physical impacts), exposure (presence of people and assets in places that could be adversely affected), and vulnerability, which is divided into sensitivity (intrinsic susceptibility to harm) and adaptive capacity (ability to cope and adapt) (Fig. 2). For these categories, variable selection was guided by the urban climate literature (Stewart and Oke, 2012) and established HVI protocols (Niu et al., 2021; Cheng et al., 2021) to ensure comprehensive coverage of physical heat mechanisms, health vulnerability, and adaptive capacity. The dataset comprised 13,170 spatiotemporal observations (2634 grids × 5 years) (See Table 1). Details of data acquisition, preprocessing, and normalization methods are provided in Appendix Table B.

Climate hazards included the intensity and frequency of physical heat events. To measure the frequency of extreme events, we utilized heat wave days (daily maximum temperature ≥ 33 °C) and tropical nights (daily minimum temperature ≥ 25 °C), while warm spell duration was used to capture the persistence of heat stress. Land surface temperatures (LSTs) derived from NASA MODIS satellite imagery were also used as a direct proxy for urban surface heat accumulation.

Urban exposure reflected the spatial distribution of populations and assets directly exposed to heat waves. For this component, the population density, building density (floor–area ratio), and impervious surface ratio, which measures the proportion of impervious surfaces that absorb and retain heat, were used. Transport accessibility was retained within Urban Exposure because high transit infrastructure density in Seoul's urban centers concentrates pedestrian and commuter populations within thermally stressed areas, thereby increasing the spatial extent of heat exposure rather than reducing it (Reid et al., 2009).

Social sensitivity represented the intrinsic characteristics that increase susceptibility to heat stress, independent of exposure levels. The sizes of the elderly (65+ years) and young children (0–4 years) populations were employed to account for physiological thermoregulatory limitations (Kenny et al., 2010), while the proportion of the population with a low income or underground housing was used as a socioeconomic indicator, capturing resource scarcity and substandard living conditions that exacerbate health risks (World Health Organization, 2024; Chakraborty et al., 2019).

Finally, adaptive capacity assessed the ability to mitigate heat impacts. Summer electricity usage per household was employed as a proxy for individual access to active cooling (e.g., air conditioning), while public infrastructure resilience was measured using the

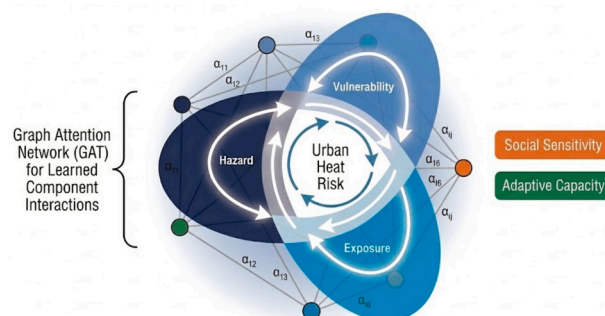


Fig. 2. Urban heat risk assessment framework adapted from IPCC AR6.

Table 1
Climate and sociodemographic variables used in the present study.

Component	Variable	Unit	Data Source Category
Climate Risk	Heat wave days	days	Korea Meteorological Administration(KMA)
	Tropical nights	nights	Korea Meteorological Administration(KMA)
	Warm spell duration	days	Korea Meteorological Administration(KMA)
	Land surface temp.	°C	Satellite imagery (MODIS)
Urban Exposure	Population density	persons/km ²	National statistics (SGIS)
	Impervious surface	%	Land cover map (EGIS)
	Building density	%	National spatial platform
	Transport accessibility	dimensionless	Public Seoul data portal
Social Sensitivity	Underground housing	%	Address-based analysis
	Young children	%	National statistics
	Low income	%	Public data portal
	Elderly population	%	National statistics
Adaptive Capacity	Summer cooling electricity	kWh/household	KEPCO + SGIS
	Healthcare accessibility	dimensionless	National spatial platform
	Green space	m ² /person	National spatial platform
	Heatwave shelter	facilities/10 k pop	National disaster data

heatwave shelter density and healthcare accessibility. Green space per capita was used to represent the city's natural cooling potential and ecosystem services (Reid et al., 2009; Melis et al., 2023; Bowler et al., 2010). Healthcare accessibility and heatwave shelter density together capture the evacuative dimension of mobility, namely the ability to reach cooling centers or medical facilities during extreme heat events. This function is conceptually distinct from transport accessibility's role in amplifying population exposure within thermally stressed urban centers, which is why the two variables are assigned to separate AR6 components.

2.3. Methods

2.3.1. Spatiotemporal graph attention network architecture

We designed a spatiotemporal GAT model to simultaneously capture spatial dependencies and temporal changes in the urban heat

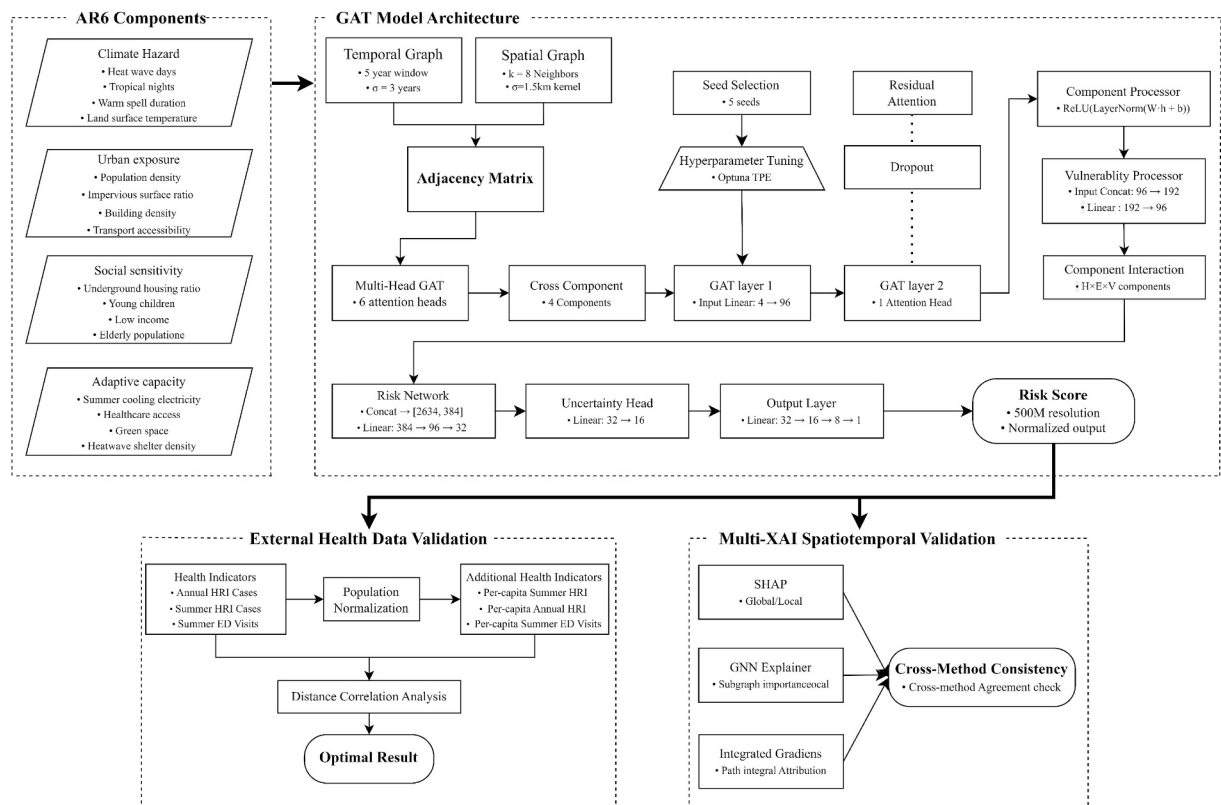


Fig. 3. Overall research framework and model architecture.

risk (Fig. 3). GATs overcome the fixed-weight limitation of conventional graph convolutional networks, which assign equal importance to all adjacent nodes, by dynamically determining connection strength between nodes based on learnable attention weights (Veličković et al., 2017; Wu et al., 2020). This enabled the differential learning of heterogeneous spatial dependencies such as the cooling effect of Han River greenery and heat amplification effects in densely populated residential areas (Jin et al., 2023; Bai et al., 2021).

We used $k = 8$ nearest neighbor connections based on LCZ scaling principles as this value corresponds to the upper bound of the LCZ spatial extent criterion (400–1500 m) at our 500 m grid resolution, yielding an approximate neighborhood radius of 1.5 km that reflects a geometric property of the grid structure rather than a kernel bandwidth parameter (Stewart and Oke, 2012). A sensitivity analysis across $k = 6, 8, 10$, and 12 was conducted to confirm that the spatial autocorrelation hierarchy among AR6 components remained invariant across alternative graph connectivity parameters (Appendix C, Table C3). Through hyperparameter optimization, we set the optimal temporal decay weight at $\sigma = 3.0$ years, balancing recent temporal dependencies with multi-year patterns.

$$w_{ij}^{spatial} = \begin{cases} \frac{1}{d_{ij} + \epsilon} & \text{if } j \in \mathcal{N}_8(i) \\ 0 & \text{otherwise} \end{cases} \quad (1a)$$

where d_{ij} is the geographic distance between nodes i and j , $\epsilon = 10^{-8}$, and $\mathcal{N}_8(i)$ denotes the $k = 8$ nearest neighbors of node i . Weights are row-normalized as $\tilde{w}_{ij} = w_{ij} / \sum_k w_{ik}$.

$$w_{ij}^{temporal} = \exp\left(-\frac{|t_i - t_j|^2}{2 \times 3.0^2}\right) \quad (1b)$$

$$w_{final} = \alpha \cdot w_{ij}^{spatial} + (1 - \alpha) \cdot w_{ij}^{temporal} \quad (2)$$

The GAT model learned the influence between nodes using LeakyReLU activation and Softmax normalization. The multi-head structure learned different attention patterns in parallel through six attention heads, capturing spatial dependencies at various scales from microscopic heat propagation to macroscopic wind corridor patterns (Vaswani et al., 2017; Chung et al., 2014; Zhao et al., 2019) (detailed equations in Appendix A).

The 16 input variables are aggregated into four AR6 components. Each component was designed to preserve unique characteristics while effectively propagating information through 96 hidden dimensions, as determined using hyperparameter optimization (Section 3.1, Table 3) and LayerNorm normalization. LayerNorm was selected due to its batch size independence and stable integration with spatiotemporal attention operations (Ba et al., 2016; Xiong et al., 2020). We also employed a two-tier dropout strategy to prevent overfitting. First, a general dropout rate (0.2301, Table 3) was applied to all layers. Component-specific dropout rates were then applied; adaptive capacity had the highest dropout rate of 0.8 due to data heterogeneity, compared to 0.6 for auxiliary components and 0.4 for the remaining components (Srivastava et al., 2014).

Component-specific processors were designed to preserve the physical meaning of the variables. Hazards, exposure, and sensitivity used ReLU for non-negative constraints, while adaptive capacity used Tanh to express mitigation and suppression effects (Raissi et al., 2019). All processors were implemented using linear, LayerNorm, activation function, and dropout stages (Karpatne et al., 2017; Willard et al., 2022). The use of Tanh for adaptive capacity was designed in consideration of maladaptation phenomena (IPCC, 2022; Tomlinson et al., 2011).

$$\text{Dropout}(\mathbf{h}, \mathbf{p}) = \begin{cases} \mathbf{p} = 0.8 & \text{if comp} = \text{adaptive} \\ \mathbf{p} = 0.6 & \text{if comp} = \text{aux} \\ \mathbf{p} = 0.4 & \text{otherwise} \end{cases} \quad (3)$$

$$\text{adaptive}_{raw} = \text{Dropout}(\text{Tanh}(\text{LayerNorm}(\text{Linear}_{96}(\mathbf{h}_{adaptive})))) \quad (4)$$

Temperature scaling was also used to adjust the influence of the regional adaptive capacity (Hinton et al., 2015). The learnable T value (initial: 1.5) was constrained to $[0.5, 2.5]$ using a clamp function, which mitigated the influence of the regional adaptive capacity in regions with complex adaptation patterns and strengthened it in clear regions based on data. The auxiliary components provided additional contextual information for the main adaptive capacity indicators by incorporating learnable latent vectors that captured unobserved regional heterogeneity, reflecting differences between regions with identical indicator values (Szegegy et al., 2015). The main and auxiliary components were combined with an optimal weight of 0.3.

$$T_{clamp} = \text{clamp}(|T|, 0.5, 2.5) \quad (5a)$$

$$\text{adaptive}_{scaled} = \tanh\left(\frac{\text{adaptive}_{raw}}{T_{clamp} + 10^{-6}}\right) \quad (5b)$$

$$\text{aux}_{processed} = \text{Dropout}\{\text{ReLU}(\text{LayerNorm}(W_{aux} \cdot \mathbf{h}_{aux} + \mathbf{b}_{aux}))\} \quad (5c)$$

$$\text{adaptive}_{final} = \text{adaptive}_{scaled} + 0.3 \times \tanh(\text{aux}_{processed}) \quad (5d)$$

To learn nonlinear interactions among the AR6 components, we implemented a cross-component attention mechanism (Fig. 3). Each AR6 component is treated as an independent, theoretically grounded module, and their multiplicative interaction effects are captured through learnable attention weights between components, enabling the model to learn component-wise amplification effects from data rather than imposing a predetermined weighting scheme. Risk was defined based on the complex interactions between hazards, exposure, and vulnerability, with nonlinear relationships emphasized when the combined effect of hazards and exposure on vulnerability exceeded simple addition. A multi-head attention structure with four heads was applied to capture these interactions in parallel across different representation subspaces (Eqs. (6a)–(6c)).

$$\text{AR6}_{\text{components}} = \text{stack}([\text{hazard}, \text{exposure}, \text{sensitivity}, \text{adaptive}]) \quad (6a)$$

$$\text{Attended}_{\text{components}} = \text{MultiHeadAttention}^{4\text{heads}}(\text{AR6}_{\text{components}}) \quad (6b)$$

$$[\text{hazard}_{\text{att}}, \text{exposure}_{\text{att}}, \text{sensitivity}_{\text{att}}, \text{adaptive}_{\text{att}}] = \text{unbind}(\text{Attended}_{\text{components}}) \quad (6c)$$

Based on the attended components in Eq. (6), we constructed a vulnerability network and a risk network. The vulnerability network implemented the Vulnerability = g(Sensitivity, Adaptive Capacity) relationship as a neural network (Eq. (7)). The risk network modeled nonlinear effects through multiplicative interaction terms among components (Eq. (8a)), combining four components and interaction terms to construct a 384-dimensional risk input (Eq. (8b)). This risk input was compressed to 96 dimensions using the f_risk function, which consisted of a two-layer neural network and residual connections, then projected onto a single scalar and normalized to a [0,1] interval to represent relative risk levels within Seoul (Eq. (8c)). Residual connections mitigated gradient vanishing (He et al., 2016), while ReLU activation ensured non-negativity constraints for risk. (See Appendix A.5 for more details of the architecture.)

$$\text{vulnerability}_{\text{input}} = \text{concat}([\text{sensitivity}_{\text{att}}, \text{adaptive}_{\text{att}}]) \quad (7)$$

$$\text{interaction} = \text{hazard}_{\text{att}} \times \text{exposure}_{\text{att}} \times \text{vulnerability} \quad (8a)$$

$$\text{risk}_{\text{input}} = \text{concat}([\text{hazard}_{\text{att}}, \text{exposure}_{\text{att}}, \text{vulnerability}, \text{interaction}]) \quad (8b)$$

$$\text{risk}_{\text{final}} = \frac{\text{risk}_{\text{scalar}} - \min(\text{risk}_{\text{scalar}})}{\max(\text{risk}_{\text{scalar}}) - \min(\text{risk}_{\text{scalar}})} \quad (8c)$$

where $\text{risk}_{\text{scalar}}$ represents the 96-dimensional risk representation projected to a single scalar value.

2.3.2. Loss functions and training

We designed a curriculum learning-based loss function to simultaneously learn data patterns, physical theory, and the graph structure via progressively adjusted weights (Bengio et al., 2009). Initially, the model had difficulty directly learning complex AR6 theoretical relationships. Therefore, we employed a two-stage weighting strategy. In the warm-up stage (the first 25 epochs), we assigned high weights to reconstruction and deep graph infomax (DGI) losses to establish the spatiotemporal patterns and graph structure. In the main stage (the remaining epochs), we progressively increased the weight on the AR6 theoretical loss to learn detailed risk interactions (Wang et al., 2021). DGI loss (Eq. (9)) maximizes mutual information between local subgraph representations around nodes and global graph vectors to preserve the graph structure (Velićković et al., 2018; Chen et al., 2020), while AR6 theoretical loss (Eq. (10)) minimizes the difference between the predicted risk and theoretical risk (Risk = Hazard × Exposure × Vulnerability) to ensure physical consistency with climate science theory. Reconstruction loss (Eq. (11)) restores the original 16 input variables from learned representations to confirm information preservation and prevent representation collapse (Vincent et al., 2008).

$$L_{\text{DGI}} = -E_{x \sim p_{\text{data}}} [\log \sigma(f(h_{\text{pos}}, s))] - E_{x \sim p_{\text{corrupt}}} [\log(1 - \sigma(f(h_{\text{neg}}, s)))] \quad (9)$$

$$L_{\text{AR6}} = \text{MSE}(\text{Risk}_{\text{predicted}}, \text{Risk}_{\text{theoretical}}) \quad (10)$$

$$L_{\text{recon}} = \text{MSE}(\text{Decoder}(\text{risk}_{\text{final}}), \text{Input}_{\text{original}}) \quad (11)$$

The final loss function adjusted the balance between these three objectives during training via curriculum-based dynamic weighting (Kendall et al., 2018; Chen et al., 2018). The mixed loss is presented in Eq. (12), where $w(t)$ decreased from 0.8 to 0.5 as training progressed to progressively shift focus from data patterns to theoretical relationships. The total loss (Eq. (13)) integrated all objectives, with mix weight(t) increasing from 0.2 to 0.9 during training, while the baseline $0.1 \times L_{\text{DGI}}$ term maintained the graph structure information across all training stages.

$$L_{\text{mixed}} = w(t) \cdot L_{\text{recon}} + (1 - w(t)) \cdot L_{\text{AR6}} \quad (12)$$

$$L_{\text{total}} = (1 - \text{mix_weight}(t)) \cdot L_{\text{DGI}} + \text{mix_weight}(t) \cdot L_{\text{mixed}} + 0.1 \cdot L_{\text{DGI}} \quad (13)$$

For hyperparameter optimization, 150 Bayesian searches were conducted using the Optuna TPE framework (Akiba et al., 2019), aiming to minimize the mean validation root mean square error (RMSE) of nested spatiotemporal cross-validation (Bergstra and

Bengio, 2012; Snoek et al., 2012).

To quantify the prediction uncertainty, we performed Monte Carlo dropout with 200 iterations to produce confidence intervals (Gal and Ghahramani, 2016). As a regularization technique, we employed spectral normalization to control the model weight scale and prevent divergence during training (Miyato et al., 2018). To avoid gradient explosion, we limited the L2 norm threshold for gradient clipping to 1.0, facilitating stable convergence (Pascanu et al., 2013). In addition, to verify the reproducibility of the model's results, we conducted experiments with five standard seeds (42, 123, 456, 789, and 999) (Reimers and Gurevych, 2017). We selected the seed with the highest performance as the final model while meeting the stability criterion (i.e., a coefficient of variation below 0.05) (Henderson et al., 2018; D'Amour et al., 2022).

2.4. Multi-XAI framework for spatiotemporal interpretation

To ensure reliable model interpretation, we comprehensively validated three XAI techniques based on heterogeneous principles: SHAP, GNNExplainer, and integrated gradients. While previous studies have typically employed a single technique (Gagne II et al., 2019; Arrieta et al., 2020), this study ensured interpretation consistency via cross-validation. We cross-validated different principles based on game theory, graph structure, and path integral (Samek et al., 2017).

SHAP was used to compute Shapley values for each grid-year pair across 2634 grids and aggregate them to 25 administrative districts to analyze risk disparities at the metropolitan scale (Eq. (14)).

$$\Phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (14)$$

where $F = \{x_1, x_2, \dots, x_{16}\}$ is the set of all 16 input features.

GNNExplainer analyzed the connection patterns between each 500 m grid and its $k = 8$ nearest neighbors at a local scale to identify subgraph structures and key variables for adjacent regions that influence heat risk. These local-scale results were aggregated to five functional regions to assess regional-scale risk patterns (Ying et al., 2019; Hamilton et al., 2017) (Eq. (15)).

$$G_{\text{important}} = \underset{G_S \subseteq G}{\operatorname{argmax}} \operatorname{Mutual Information}(Y, f(G_S)) \quad (15)$$

Integrated gradients computed variable contributions at the grid level through path integration from baseline ($\bar{x}$) to actual value (x) to capture individual grid-level risk factors. These grid-level results were also aggregated to the regional scale for a comparison with GNNExplainer (Eq. (16)).

$$\operatorname{IG}_i(x) = (x_i - \bar{x}_i) \times \int_0^1 \frac{\partial f(\bar{x} + \alpha(x - \bar{x}))}{\partial x_i} d\alpha \quad (16)$$

To investigate the consistency of the three techniques, we converted the absolute importance scores for each of the 16 variables produced by the XAI techniques into rankings (Eq. (17)) (Samek et al., 2021). We then quantified the rank differences between methods for the same variable using Spearman's rank correlation (Ribeiro et al., 2016; Adadi and Berrada, 2018).

$$d_i = \operatorname{Rank}_{\text{method1}}(i) - \operatorname{Rank}_{\text{method2}}(i) \quad (17a)$$

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (17b)$$

Additionally, spatiotemporal stability was assessed using SHAP, which produces continuous scores that are suitable for correlation analysis (Rudin, 2019; Doshi-Velez and Kim, 2017; Miller, 2019). In contrast, the discrete subgraph structures of GNNExplainer and baseline-dependent path values of integrated gradients limit direct temporal and spatial comparisons. We analyzed correlations between SHAP values across the 2015–2019 period and five functional regions to assess the temporal and spatial consistency of the interpretations, while regional variation in variable importance and inter-year rankings were examined to confirm that the XAI techniques accurately reflected regional heterogeneity and temporal stability (Arrieta et al., 2020; Bhatt et al., 2020; Confalonieri et al., 2021).

Finally, to formally assess the statistical significance of the scale-dependent transition reported in Section 3.3, bootstrap confidence intervals (10,000 iterations; Efron and Tibshirani, 1993) were computed for each AR6 component's proportional contribution across the five functional regions, and one-sample t -tests were applied to compare regional estimates against the metropolitan point estimates derived from SHAP analysis. Binomial tests and Kruskal-Wallis tests (Kruskal and Wallis, 1952) were further applied to evaluate rank consistency across regions and inter-regional distributional differences in component importance, respectively.

2.5. Cross-validation framework: spatiotemporal and external health data validation

The relationship between urban heat risk and health impacts is complex and nonlinear due to threshold effects, cumulative exposure, and nonlinear responses among vulnerable populations (Liu et al., 2025; Burkart et al., 2021). Linear correlation coefficients fail to capture these composite patterns and are particularly unsuitable for validating spatial clustering and temporal lag effects learned by the spatiotemporal attention of the GAT model. Therefore, this study adopted distance correlation, which can detect nonlinear

relationships (Székely et al., 2007), and combined it with spatiotemporal cross-validation to assess the reliability of the GAT model.

We collected six health indicators at the 25-district level for the 2015–2019 period from the National Health Insurance Service (NHIS), Korea Disease Control and Prevention Agency (KDCA), and Ministry of the Interior and Safety (MOIS) (Table 2). Three population-normalized indicators (per-capita summer heat-related illness [HRI], per-capita annual HRI, and per-capita summer emergency department [ED] visits) were used to control for population size to reflect regional vulnerability and acute impacts, while three absolute indicators (summer HRI cases, annual HRI cases, and summer ED visits) represented the healthcare burden and cumulative effects (Bao et al., 2016; Anderson and Bell, 2009). The 500 m grid predictions from the model were aggregated at the district level and compared with these health data.

Distance correlation computes distance matrices for two variables with double centering (Eq. (18)):

$$\text{Distance Matrix : } A_{ij} = |X_i - X_j|, B_{ij} = |Y_i - Y_j| \quad (18a)$$

$$\text{Double Centering : } \tilde{A}_{ij} = A_{ij} - \bar{A}_{i.} - \bar{A}_{.j} + \bar{A}_{..}, \tilde{B}_{ij} = B_{ij} - \bar{B}_{i.} - \bar{B}_{.j} + \bar{B}_{..} \quad (18b)$$

$$\text{Distance Covariance : } \text{dCov}^2(X, Y) = \frac{1}{n^2} \sum_{i,j=1}^n \tilde{A}_{ij} \tilde{B}_{ij} \quad (18c)$$

$$\text{Distance Correlation : } \text{dCor}(X, Y) = \frac{\text{dCov}(X, Y)}{\sqrt{\text{dCov}(X, X) \text{dCov}(Y, Y)}} \quad (18d)$$

This detects nonlinear relationships not captured by Pearson's correlation and effectively captures the nonlinear relationship between the complex spatiotemporal patterns of the GAT model and health outcomes.

We examined time lags (0–3 years) between heat exposure and health impacts to align with the model's temporal kernel ($\sigma = 3.0$ years), testing whether GAT-learned temporal dependencies reflected actual health impact structures (Eq. (19)). Acute illnesses occur immediately (lag 0), while cumulative effects emerge over time (lag 1–3 years) (Son et al., 2012; Vicedo-Cabrera et al., 2021). Statistical significance was assessed using permutation testing and bootstrap confidence intervals.

$$\text{Health}_{\text{lag}}(t) = f(\text{GAT}_{\text{risk}}(t - \text{lag}), \text{Confounders}(t)) \text{ where } \text{lag} \in \{0, 1, 2, 3\} \quad (19)$$

To verify the spatiotemporal interpretability of the model, we combined three systematic cross-validation strategies. First, we employed temporal forward-chaining cross-validation to prevent temporal information leakage (Bergmeir and Benítez, 2012). We progressively extended the training period (e.g., predicting 2017 using 2015–2016 data), ensuring that future information was not incorporated into training (Eq. (20)) (Tashman, 2000; Hyndman and Athanasopoulos, 2018; Harvey, 1990).

$$CV_{\text{temporal}} = \frac{1}{T - t_0} \sum_{t=t_0+1}^T \text{Performance}(M_{\text{train}}(1 : t - 1), \text{Data}_{\text{test}}(t)) \quad (20)$$

where $t \in \{2015, 2016, 2017, 2018, 2019\}$.

Second, to prevent spatial information leakage, we employed spatial block cross-validation with a leave-one-region-out approach, sequentially excluding Seoul's major administrative districts to ensure spatial independence between the training and test sets (Eq. (21)) (Roberts et al., 2017; Ploton et al., 2020; Le Rest et al., 2014).

$$CV_{\text{spatial}} = \frac{1}{R} \sum_{r=1}^R \text{Performance}(M_{\text{train}}(\text{Regions}), \text{Data}_{\text{test}}(r)) \quad (21)$$

Third, we implemented nested cross-validation to eliminate the selection bias that can occur during hyperparameter tuning (Varma and Simon, 2006). We completely separated the northeast region from the five regions as test data, trained the model using 2015–2018 data from the remaining four regions, and evaluated the spatiotemporal extrapolation performance by predicting northeast region data for 2019 (Eq. (22)). We repeated this process for all region-year combinations to verify whether the model had robust predictive capabilities for both new regions and future time points not learned in the past (Cawley and Talbot, 2010).

Table 2

External health data used for validation.

Health Indicator	Data Source	Definition	Temporal Coverage	Spatial Unit
Summer HRI Cases	NHIS	HRI cases (T67.*) in summer	2015–2019	District (25 units)
Annual HRI Cases	NHIS	Total annual HRI cases	2015–2019	District
Summer ED Visits	KDCA	ED visits in summer	2015–2019	District
Per-capita Summer HRI	NHIS + MOIS	Population-normalized summer HRI rate	2015–2019	District
Per-capita Annual HRI	NHIS + MOIS	Population-normalized annual HRI rate	2015–2019	District
Per-capita Summer ED	KDCA + MOIS	Population-normalized summer ED rate	2015–2019	District

Note: HRI = heat-related illness; ED = emergency department. All indicators were aggregated at the administrative district (Si-Gun-Gu) level. Summer HRI Cases (May 15–Sep. 30) and annual HRI cases (Jan.–Dec.) were defined as a T67.* primary diagnosis from the KDCA and NHIS, respectively. Summer ED visits (June–Sep.) from NEDIS included all chief complaints. Per-capita indicators were normalized using KOSIS population data.

$$CV_{spatiotemporal} = \frac{1}{R \times T} \sum_{r=1}^R \sum_{t=1}^T R^2(y_{r,t}, \hat{y}_{r,t}) \quad (22)$$

To complement these validation strategies, we employed global Moran's I to verify the statistical significance of the spatial patterns. Moran's I quantifies spatial autocorrelation to confirm whether observed spatial clustering represents systematic patterns rather than random distribution. This was used to validate that the spatial dependencies learned by the GAT model reflected actual risk structures and to determine how the spatial autocorrelations of individual AR6 components were integrated into the final risk patterns.

3. Results

3.1. GAT model performance and robustness validation

Optimal hyperparameters were determined using 150 Bayesian trials (Table 3). Model stability was validated with five random seeds (42, 123, 456, 789, and 999), yielding mean spatiotemporal cross-validation scores of 0.635 ± 0.013 with a coefficient of variation of 0.0205. This low variability illustrated the robustness of the proposed architecture regardless of initialization, confirming the consistent learning of spatiotemporal attention patterns. This seed-level stability, combined with the invariant spatial autocorrelation hierarchy observed across $k = 6, 8, 10$, and 12 (Appendix C, Table C3), supports the robustness of the scale-dependent transition finding to alternative parameter configurations.

In testing the generalization ability of the model with unobserved spatiotemporal data, temporal forward cross-validation and spatial block cross-validation both exhibited a high stability with an average of 0.9681, indicating no region-specific overfitting (Table 4). The average for nested spatiotemporal cross-validation was slightly higher at 0.9682, suggesting that the model produced robust predictions with unobserved spatiotemporal data.

Internal consistency verification confirmed that the model learned the IPCC AR6 framework as intended. The GAT attention mechanism amplified the input variance of all components by an average of 1.98 times, learning even subtle signals. Sensitivity experienced the strongest amplification (2.15-fold), indicating focused learning on vulnerable population characteristics. Similarly, in the component interaction analysis, the interaction between sensitivity and adaptive capacity was highest at 2.079, confirming the importance of adaptive capacity gaps. The correlation between exposure and sensitivity decreased by 93.7% from 0.804 before learning to 0.051 after learning, confirming that GAT independently separated the unique meaning of each component (Table 5).

3.2. Spatiotemporal risk assessment and spatial pattern analysis

Spatiotemporal heat risk assessment for the 2634 grids in Seoul from 2015 to 2019 revealed that the overall mean risk was 0.441 (standard deviation [SD] = 0.129). The high-resolution 500 m grid map revealed clear spatial inequality patterns, with areas of high risk concentrated in central Seoul and southeast areas, while regions of low risk were located primarily in northern mountainous and peripheral areas (Fig. 4). Central Seoul (Jung-gu and Jongno-gu) had the highest mean risk of 0.468 ($n = 1385$), compared to 0.452 for the northwest ($n = 1515$), 0.447 for the northeast ($n = 3665$), 0.429 for the southeast ($n = 3050$), and 0.429 for the southwest ($n = 3555$). Appendix C presents more detailed regional statistics.

Global Moran's I analysis confirmed strong positive spatial autocorrelation (Moran's $I = 0.4978$, Z-score = 51.64, $p < 0.001$) for the final risk scores (Table 6), indicating that grid cells with similar risk levels were clustered systematically rather than randomly. Of the individual components, climate hazards had the highest Moran's $I = 0.9718$, confirming the strong spatial concentration of climate phenomena, while the final risk (0.4978) represented the nonlinear integration of heterogeneous spatial patterns for the four

Table 3
GAT model hyperparameter optimization results and architecture configuration.

Parameter Category	Parameter	Search Space	Optimal Value
Architecture	hidden_dim	[32, 64, 96, 128]	96
	gat_heads	[4, 6, 8]	6
	dropout	[0.1, 0.5]	0.2301
Learning	learning_rate	[1e-4, 1e-2]	7.036e-4
	batch_size	[2, 4, 8]	4
Spatiotemporal Fusion	spatiotemporal_alpha	[0.3, 0.8]	0.5829
Training Control	training_epochs	Fixed	250
	patience	Fixed	50
	warmup_epochs	Fixed	30
Attention Mechanism	adaptive_temperature	–	1.6133
Search Configuration	total_trials	–	150
	best_trial	–	Trial 47/150
Reproducibility	tested_seeds	[42, 123, 456, 789, 999]	456*
(Five-seed validation)	mean_performance $\pm$ std	–	0.635 ± 0.013
	coefficient_of_variation	–	0.0205

Note: *Selected as the optimal seed based on the highest performance (0.6515) and realistic spatial pattern (spatial autocorrelation = 0.2474). Dropout (0.2301) represents the general dropout rate applied to all layers. Component-specific dropout rates (Eq. (3)) were also employed.

Table 4
Model validation results.

Validation Method	Configuration	Score	RMSE
Temporal Forward Cross-Validation	2015–2016 → 2017	0.9682	0.161
	2015–2017 → 2018	0.9681	0.162
	2015–2018 → 2019	0.9681	0.162
Spatial Block Cross-Validation	5-fold regional	0.9681	0.163
	Region 0 (624 pts./yr)	0.9689	0.159
	Region 1 (580 pts./yr)	0.9669	0.164
	Region 2 (684 pts./yr)	0.9691	0.158
	Region 3 (746 pts./yr)	0.9675	0.162
Nested Spatiotemporal	20 region–year combinations	0.9682	0.161
	LORYO (most strict)	0.9682	0.161

Table 5
Validation of the internal consistency of the GAT model.

Analysis Component	Metric	Value
GAT Attention Enhancement		
Hazard variance	Input → Output	0.0662 → 0.1186 (1.79×)
Exposure variance	Input → Output	0.0594 → 0.1173 (1.98×)
Sensitivity variance	Input → Output	0.0289 → 0.0620 (2.15×)
Adaptive variance	Input → Output	0.0216 → 0.0434 (2.01×)
AR6 Component Interactions		
Sensitivity ↔ Adaptive	–	2.079
Exposure ↔ Sensitivity	–	2.062
Hazards ↔ Sensitivity	–	1.970
Exposure ↔ Adaptive	–	1.993
Hazards ↔ Sensitivity	–	1.970
Hazards ↔ Adaptive	–	1.901
Multicollinearity Resolution		
Hazards ↔ Exposure	Before → After	0.272 → 0.018 (–93.4%)
Hazards ↔ Sensitivity	Before → After	0.216 → 0.069 (–68.1%)
Exposure ↔ Sensitivity	Before → After	0.804 → 0.051 (–93.7%)
Sensitivity ↔ Adaptive	Before → After	0.046 → 0.004 (–91.3%)

Note: The GAT attention mechanism amplified the input variance (avg. 1.98×) while reducing multicollinearity. The variance inflation factor for adaptive capacity decreased from >10.0 to 1.057 due to enhanced dropout and layer normalization.

components.

Comparing the 500 m grid resolution with administrative aggregation revealed substantial spatial heterogeneity masked by district-level averaging (Fig. 4). Within individual districts, grids adjacent to green spaces and high-density commercial zones exhibited markedly different risk levels, demonstrating that administrative boundaries failed to capture the fine-scale spatial variability of urban heat risk. The 500 m resolution thus effectively identified localized high-risk hotspots that were obscured in the coarser aggregation scheme.

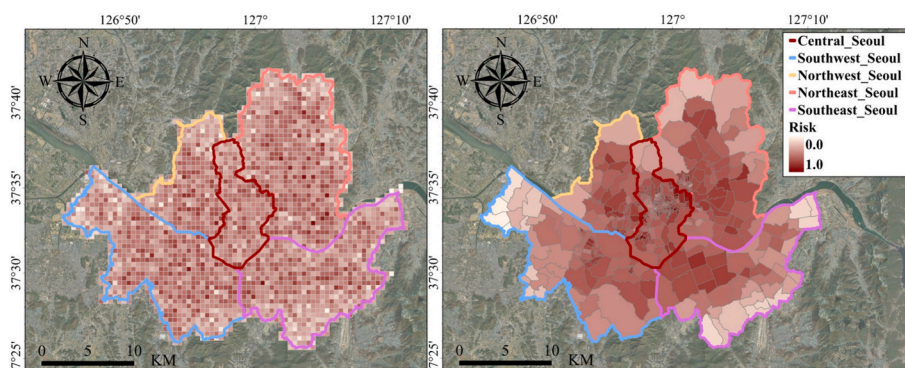


Fig. 4. Multi-scale heat risk maps: (left) 500 m grid-level and (right) district-level.

Table 6
Spatial autocorrelation analysis of the AR6 components.

Component	Moran's I	Z-score	p-value
Climate Hazards	0.9718	100.76	<0.001
Social Sensitivity	0.8528	88.42	<0.001
Urban Exposure	0.7378	76.51	<0.001
Adaptive Capacity	0.5627	58.36	<0.001
Overall Risk	0.4978	51.64	<0.001

3.3. Multi-XAI interpretation results

The three heterogeneous XAI techniques together formed a strong consensus in terms of model interpretation. At the metropolitan scale, SHAP analysis across all Seoul grids (13,170 grid-year observations aggregated to 25 districts over 5 years) revealed that urban exposure had the highest component contribution at 33.73%, followed by social sensitivity (28.23%), climate hazards (23.57%), and adaptive capacity (14.47%). Pairwise correlations between methods ranged from $\rho = 0.82$ to 0.89 (all $p < 0.001$), with 89.6% agreement in top-five variable rankings (See Table 7).

At the individual variable level, the three XAI methods showed high consistency in identifying critical risk factors (Table 8). Impervious surface area, elderly population, and building density consistently ranked as the top-three variables, with consensus rates exceeding 90%, and the Spearman rank correlation between SHAP and GNNExplainer was $\rho = 0.89$ ($p < 0.001$). Within this high-agreement framework, inter-method disagreement was most pronounced in the Climate Hazard component based on GNN – SHAP rank differences (mean = 5.75), consistent with the spatial autocorrelation hierarchy in Table 6. The strong spatial clustering of Climate Hazard (Moran's $I = 0.972$) amplified discrepancies between GNNExplainer, which operates on local graph neighborhoods, and SHAP, which aggregates global contributions. Adaptive Capacity showed the greatest discrepancy relative to Integrated Gradients (mean $|IG - SHAP| = 3.25$), reflecting the sensitivity of path-integral attribution to baseline selection. Among individual variables, heat wave days and cooling electricity each showed the largest discrepancies by GNN – SHAP and IG – SHAP criteria, respectively ($|diff| = 9$; Appendix C, Table C4).

Temporal consistency analysis confirmed that importance rankings remained stable for major factors (urban exposure and social sensitivity: $\rho > 0.85$ from 2015 to 2019), while climate hazards showed a gradual increase ($\rho: 0.82 \rightarrow 0.89$), capturing both structural patterns and climate variability (Fig. 5).

However, when the analysis scale shifted from metropolitan (SHAP district aggregation) to regional (GNNExplainer and integrated gradients aggregated to five functional regions), a statistically confirmed transition in dominant risk factors was observed. While urban exposure (33.73%) dominated at the metropolitan scale, climate hazards contributed 62–68% across all regions at the regional scale (Table 9, Fig. 6), with bootstrap confidence intervals showing no overlap between scales for all four AR6 components (see Table 9 note). This scale-dependent transition indicates that metropolitan-scale analysis captures between-district structural disparities, whereas regional-scale analysis reveals within-region climate exposure mechanisms.

3.4. External health data validation: distance correlation

GAT model-predicted risk had statistically significant associations with actual health impacts. Per-capita summer HRI showed the highest correlation, with a distance correlation of 0.449 ($p < 0.001$), followed by per-capita annual HRI (0.422, $p < 0.001$) and per-capita summer ED visits (0.397, $p < 0.001$) (Table 10, Fig. 7). Four of the six indicators were significant at $p < 0.05$, supporting the empirical validity of the model for comparative risk prioritization at the district scale, with associations notably exceeding those reported for conventional HVI approaches (Niu et al., 2021).

Time lag analysis (0–3 years) revealed that population-adjusted indicators had the highest correlations without a time lag (mean 0.423, $p < 0.001$), demonstrating that the GAT model immediately reflected acute health impacts in the current year. In contrast, the absolute value indicators exhibited correlations with a lag of 2 or 3 years, reflecting the long-term healthcare system burden. Statistical robustness was verified based on bootstrap confidence intervals (1000 iterations) and permutation testing.

Table 7
Cross-method consistency metrics across multiple analytical dimensions.

Analysis Method	Analysis Scope	Methods Used	Consistency	Spearman ρ	p-value
Metropolitan-scale Importance	13,170 points	SHAP + GNNExplainer + IG	89.6%	0.89	<0.001
Feature Ranking	16 AR6 variables	All methods	95.0%	0.85	<0.001
Component Hierarchy	4 AR6 components	All methods	100.0%	0.92	<0.001
Temporal Consistency	2015–2019 (5 years)	SHAP + IG	85.0%	0.81	<0.001
Temporal Trends	5-year progression	SHAP + IG	90.0%	0.83	<0.001
Spatial Patterns	5 Seoul regions	GNNExplainer + IG	80.0%	0.78	<0.001
Regional Analysis	District-level	GNNExplainer + IG	85.0%	0.80	<0.001
SHAP $\leftrightarrow$ GNNExplainer	13,170 points	–	–	0.87	< 0.001
SHAP $\leftrightarrow$ Integrated Gradients	13,170 points	–	–	0.85	< 0.001
GNNExplainer $\leftrightarrow$ IG	13,170 points	–	–	0.82	< 0.001

Table 8
Metropolitan-scale variable importance rankings with multi-XAI consensus rates.

Component	Variable	Mean Importance	Consensus (%)
Urban Exposure	–	33.73%	–
Social Sensitivity	–	28.23%	–
Climate Hazards	–	23.57%	–
Adaptive Capacity	–	14.47%	–
Exposure	Impervious surface	0.368 ± 0.014	94.2
Sensitivity	Elderly population	0.341 ± 0.010	92.8
Exposure	Building density	0.322 ± 0.006	91.3
Exposure	Population density	0.319 ± 0.004	90.7
Hazards	Land surface temperature	0.282 ± 0.006	89.2
Sensitivity	Low income	0.270 ± 0.005	88.5
Exposure	Transport accessibility	0.213 ± 0.005	86.3
Sensitivity	Underground housing	0.208 ± 0.004	85.7
Sensitivity	Young children	0.205 ± 0.004	85.1
Hazards	Heat wave days	0.200 ± 0.004	84.6

Note: Consensus represents the agreement between SHAP, GNNExplainer, and integrated gradients for the top-three variable rankings. Mean importance was derived from SHAP values at the metropolitan scale.

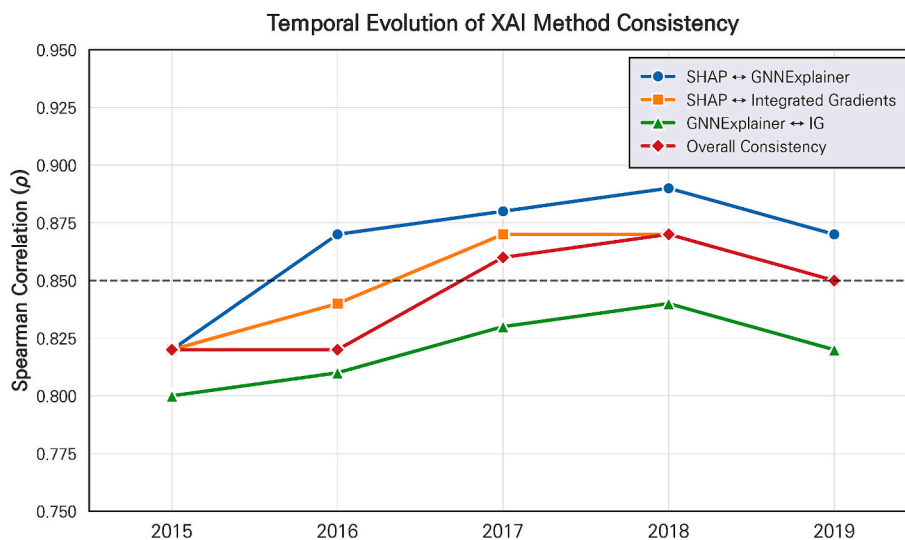


Fig. 5. Temporal stability of XAI-derived importance rankings (2015–2019).

Table 9
Regional-scale component heterogeneity analysis.

Functional Region	Risk	Rank	Dominant Component	Contribution (%)	Secondary Component	Contribution (%)
Central Seoul	0.4676	1	Climate hazards	67.0	Urban exposure	18.1
Northwest Seoul	0.4517	2	Climate hazards	66.1	Urban exposure	17.7
Northeast Seoul	0.4472	3	Climate hazards	62.2	Urban exposure	17.8
Southeast Seoul	0.4291	4	Climate hazards	66.5	Adaptive capacity	12.0
Southwest Seoul	0.4290	5	Climate hazards	68.2	Urban exposure	17.0

Note: Regional-scale analysis aggregated GNNExplainer results (operating at local scale with 500 m grids and $k = 8$ neighbors) and integrated gradients results (operating at grid level) to five functional regions. All five regions were dominated by climate hazards (62–68%) at the regional scale, while urban exposure was dominant (33.73%) at the metropolitan scale. The scale-dependent shift was statistically confirmed: mean regional climate hazard importance (66.0%, 95% CI: 64.0–67.5%) was entirely non-overlapping with the metropolitan estimate (one-sample t -test: $t = 42.32$, $p < 0.001$; see Section 2.4 for bootstrap procedure). Climate hazards ranked first in all five regions without exception (binomial test: $p < 0.001$), and component importance distributions differed significantly across regions (Kruskal-Wallis: $H = 16.14$, $p = 0.001$).

4. Discussion

The present study found that the dominant factors determining urban heat risk fundamentally differ depending on the scale of analysis. This is in line with previous studies that have reported that ecological and climate systems exhibit scale-dependent dynamics,

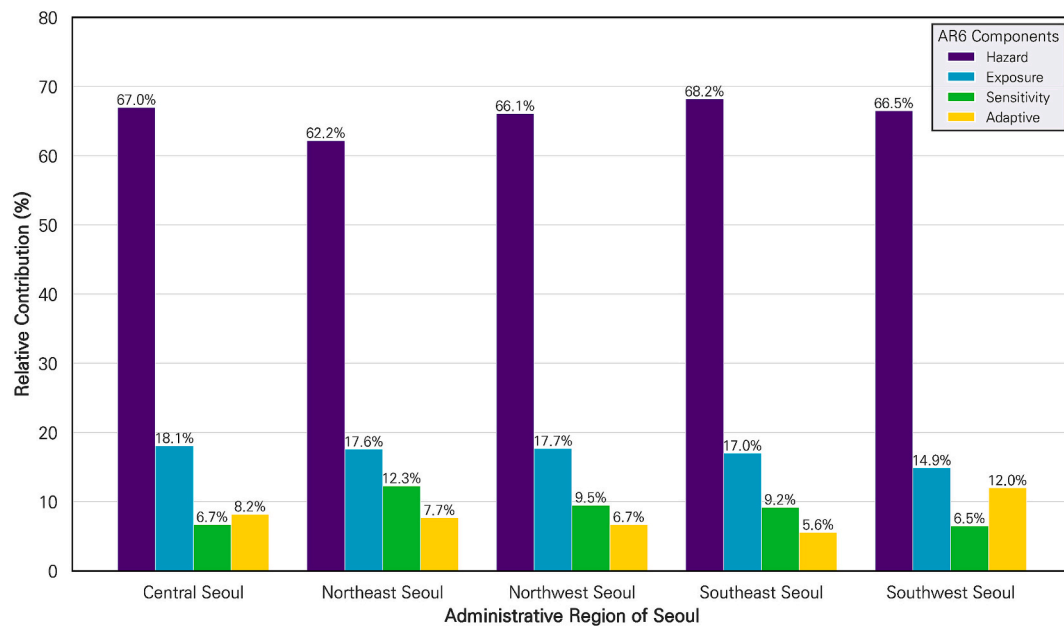


Fig. 6. Component contributions at regional scale across Seoul's five functional regions.

Table 10

GAT model external validation using distance correlation.

Rank	Health Indicator	Distance Correlation	95% CI	<i>p</i> -value*	Lag (Years)	Coverage Period
1	Per-capita Summer HRI	0.449	[0.314, 0.605]	<0.001***	0	2015–2019
2	Per-capita Annual HRI	0.422	[0.293, 0.598]	<0.001***	0	2015–2019
3	Per-capita Summer ED Visits	0.397	[0.304, 0.532]	<0.001***	0	2015–2019
4	Annual HRI Cases	0.321	[0.250, 0.509]	0.058	3	2015–2019
5	Summer HRI Cases	0.316	[0.233, 0.512]	0.026*	2	2015–2019
6	Summer ED Visits	0.219	[0.215, 0.455]	0.427	3	2015–2019

Abbreviations: HRI = Heat-related illness; ED = Emergency department. Data sources: NHIS = National Health Insurance Service; KDCA = Korea Disease Control and Prevention Agency; KOSIS = Korean Statistical Information Service. Note: *Statistically significant based on permutation testing (1000 iterations). *p*-values: ****p* < 0.001, ***p* < 0.01, **p* < 0.05.

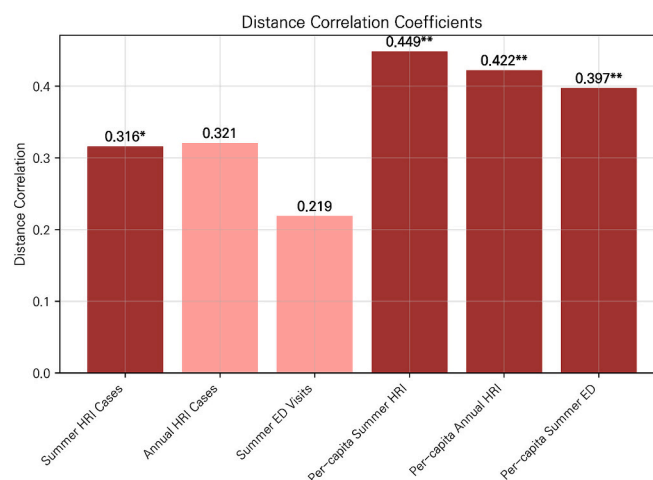


Fig. 7. Distance correlation between GAT-predicted risk and heat-related health outcomes across six indicators.

where dominant drivers shift with observational grain size (Lara et al., 2020; Huang et al., 2023). This study empirically demonstrated this scale dependency, with urban exposure the primary driver of risk at the metropolitan scale and climate hazards dominating at the regional scale. Changes in spatial autocorrelation provided mechanistic support for this transition. As the aggregation unit increases in size, local variation is averaged out, revealing structural patterns (Fotheringham and Wong, 1991). Conversely, at regional and local scales, the physical patterns associated with climate hazards combine with heterogeneous distributions of socioeconomic factors to weaken spatial continuity. This suggests that urban climate risk is not determined solely by spatial proximity to neighboring areas, but that social context fundamentally affects the spatial structure.

The dominance of urban exposure at the metropolitan scale can be attributed to the persistence of spatial structures that became entrenched during Seoul's compressed growth process. Historical development patterns concentrated infrastructure investment in specific districts, creating persistent disparities in impervious surface coverage, building density, and green space accessibility (Bae and Joo, 2020). With these initial development decisions constraining subsequent urban structural evolution, structural inequalities have accumulated and cannot easily be resolved through short-term policy interventions (Martin and Sunley, 2006). The scale-dependent risk identified in the present study suggests that, while historically entrenched structural inequality has a strong effect on metropolitan-level risk patterns, regional variation in climate hazards is more important at the regional scale. This suggests that determining which areas are most vulnerable and explaining why they are vulnerable must be addressed at fundamentally different spatial scales.

Existing vulnerability indices generally fail to capture this scale transition due to methodological limitations. For example, principal-component-based linear aggregation approaches overlook nonlinear interactions among urban components, leading to low correlations between HVIs and actual health outcomes (Niu et al., 2021; Conlon et al., 2020). In contrast, our GAT model dynamically learned spatial dependencies such as the propagation of cooling effects from green spaces or heat accumulation in high-density developments through attention mechanisms, accurately capturing the interactions among AR6 components (Klemmer et al., 2023). The methodological validity of this approach was confirmed using health data, with the immediate effects of population-adjusted indicators reflecting socioeconomic vulnerability and the lagged effects of absolute value indicators capturing the temporal accumulation of heat exposure (Bao et al., 2016; Anderson and Bell, 2009). By demonstrating meaningful associations with actual health outcomes at the district scale, our approach supports the development of spatially differentiated adaptation strategies and advances the case for health-outcome-linked risk assessment, moving beyond geographic classification toward empirically grounded vulnerability prioritization.

Because of the effects of the analysis scale on heat risk assessment, a dual-track policy approach is required. At the metropolitan scale, policies should address structural inequality through the equitable distribution of cooling infrastructure between districts. At the regional scale, interventions should target climate hazard hotspots using localized measures such as cool pavement, urban greening, and enhanced shelter networks. Effective climate adaptation also requires vertical integration within multilevel governance systems. Regional governments should be responsible for long-term strategic planning and equitable resource allocation, while local governments should be in charge of context-specific actions (Keskitalo et al., 2016). Short-term adaptation without structural improvement risks reproducing inequality, while long-term planning without immediate protection would neglect currently vulnerable populations.

Theoretically, this study introduces the spatial scale to climate justice discourse. While previous research has distinguished distributive and procedural justice using capability approaches (Schlosberg, 2012), discussion of the scales at which these operate has been lacking. Our analysis revealed clear inequalities in risk exposure among vulnerable populations at the metropolitan scale; however, at the regional scale, the uneven distribution of risk mitigation infrastructure resulted in high-income groups facing similar risks, generating a complex pattern (Carvalho et al., 2022). This suggests that climate justice cannot be reduced to a single principle, and that different concepts of justice must be considered depending on the scale.

Nevertheless, this study has three major limitations. First, the district-level health data were spatially mismatched with the 500 m grid predictions, limiting direct verification of microscale risk patterns. This spatial aggregation attenuates within-district heterogeneity, suggesting that the observed distance correlation of 0.449 likely represents a conservative lower bound, which future studies incorporating patient-geocoded clinical records could address. Second, the five-year study period (2015–2019) may underestimate the growing regional dominance of climate hazards as heat extremes intensify, a limitation that scenario-based climate projections could directly address. Third, generalizability to other cities should be approached cautiously given Seoul's unique compressed growth and high-density patterns. Future research should verify scale effects in multi-city comparisons, integrate climate scenarios for adaptation pathways, evaluate policy interventions using causal inference, and expand toward multidimensional compound hazard assessment.

5. Conclusion

This study developed a high-resolution explainable spatiotemporal GAT based on the IPCC AR6 framework. By overcoming the methodological limitations associated with conventional urban heat risk assessment, this framework represents a scientific tool for realizing climate justice. Using spatiotemporal cross-validation ($R^2 = 0.968$), multi-XAI consistency (Spearman $\rho = 0.89$), and health data validation (distance correlation = 0.449, $p < 0.001$), the predictive performance, interpretive reliability, and empirical validity of the model were verified. The key finding was that the most dominant factor affecting the urban heat risk differed according to the analysis scale. Urban exposure (33.73%) was the primary determinant of this risk at the metropolitan scale, while climate hazards (62–68%) were dominant at the regional scale. This scale-dependent transition fundamentally challenges existing single-scale HVI approaches that assume uniform risk drivers between spatial hierarchies. Spatial analysis revealed that the strong spatial concentration of climate phenomena (Moran's $I = 0.972$) transformed into compound risk patterns (0.498) through nonlinear socioeconomic interactions, proving that effective climate justice policy cannot rely on metropolitan-scale interventions alone. The ability of fine-

resolution grids to capture within-district heterogeneity that is overlooked by administrative aggregation establishes an empirical foundation for spatially differentiated adaptation strategies targeting local vulnerability hotspots. In particular, metropolitan-scale structural improvement and regional/local-scale adaptation measures must operate simultaneously, establishing the need for multi-layered approaches in climate justice policy. While limited to Seoul, adherence to the IPCC AR6 framework ensures that the proposed framework can be transferred to other megacities. Multi-city comparisons, long-term time-series analysis, causal inference integration, and extension to compound disasters can subsequently be used to transform this framework into an even more sophisticated decision-making tool for urban climate adaptation.

CRedit authorship contribution statement

Heedo Choi: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jee Soo Park:** Writing – review & editing, Project administration, Investigation, Data curation. **Chul-Hee Lim:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. GAT architecture and mathematical formulations

A.1. Attention mechanism

The attention score between node i and j in the GAT is calculated as follows:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]) \quad (\text{A1})$$

where $\mathbf{a}^T$ denotes the learnable attention vector, $\mathbf{W}\mathbf{h}_i$ is the shared linear transformation matrix, and $\parallel$ represents the concatenation operation. LeakyReLU is employed to provide training stability and capture asymmetric influence (A.1). These scores are normalized using the Softmax function to compute the final attention weights α_{ij} (A.2). The final node representation is updated via a weighted sum of neighbor nodes (A.3), where σ is the nonlinear activation function and $\mathbf{N}_i$ represents the set of neighbors for node i .

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathbf{N}_i} \exp(e_{ik})} \quad (\text{A2})$$

$$\mathbf{h}_i = \sigma \left(\sum_{j \in \mathbf{N}_i} \alpha_{ij} \mathbf{W}\mathbf{h}_j \right) \quad (\text{A3})$$

A.2. Multi-head attention

To capture diverse spatial patterns, we employ $k = 6$ attention heads in parallel (A.4). Each head k learns independent attention weights $\alpha_{ij}^{(k)}$ and transformation matrices $\mathbf{W}^{(k)}$.

$$\mathbf{h}_i^{(l+1)} = \parallel_{k=1}^6 \sigma \left(\sum_{j \in \mathbf{N}_i} \alpha_{ij}^{(k)} \mathbf{W}^{(k)} \mathbf{h}_j^{(l)} \right) \quad (\text{A4})$$

A.3. Component-specific processing

Each AR6 component undergoes two-stage GAT processing. The first stage uses six attention heads to learn various spatial patterns, while the second stage consolidates them with a single head (A.5).

$$h_1^{(comp)} = \text{ReLU}(\text{LayerNorm}(\text{GAT}_1^{6heads}(x^{(comp)}, E))) \quad (\text{A5a})$$

$$h_2^{(comp)} = \text{LayerNorm}(\text{GAT}_2^{1head}(h_1^{(comp)}, E)) \quad (\text{A5b})$$

The final component representation, incorporating residual connections, is computed as follows (A.6):

$$h_{comp} = \text{ReLU}(h_2^{(comp)} + W_{proj} \cdot x^{(comp)}) \quad (\text{A6})$$

A.4. Vulnerability network structure

The vulnerability module expands the $2 \times \text{hidden_dim}$ input to 192 dimensions, then compresses it back to 96 dimensions (A.7).

$$\text{vulnerability} = \text{Tanh}(\text{LayerNorm}(\text{Linear}_{96}(\text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{Linear}_{192}(\text{vulnerability}_{input}))))))) \quad (\text{A7})$$

A.5. Risk network structure

The risk network receives a $4 \times \text{hidden_dim}$ input and processes it through a two-stage neural network. The 384-dimensional risk input is compressed to 192 dimensions, then reduced to 96 dimensions to learn complex nonlinear relationships, with LayerNorm and ReLU activation applied at each stage (A.8). To mitigate the vanishing gradient problem, residual connections are applied (A.9). The risk score is computed by summing the hidden representation and the residual, followed by ReLU activation to ensure the non-negative constraint of physical risk (A.10).

$$\text{risk}_{hidden} = \text{ReLU}(\text{LayerNorm}(\text{Linear}_{96}(\text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{Linear}_{192}(\text{risk}_{input}))))))) \quad (\text{A8})$$

$$\text{risk}_{residual} = \text{Linear}_{96}(\text{risk}_{hidden}) \quad (\text{A9})$$

$$\text{risk}_{combined} = \text{ReLU}(\text{risk}_{hidden} + \text{risk}_{residual}) \quad (\text{A10})$$

The 96-dimensional representation is projected to a single scalar through a linear transformation (A.11).

$$\text{risk}_{scalar} = W_r \cdot \text{risk}_{combined} + b_r \quad (\text{A11})$$

The final risk score is normalized to the [0, 1] interval across all grids in Seoul (A.12).

$$\text{risk}_{final} = \frac{\text{risk}_{scalar} - \min(\text{risk}_{scalar})}{\max(\text{risk}_{scalar}) - \min(\text{risk}_{scalar})} \quad (\text{A12})$$

where 0 indicates minimum risk and 1 indicates maximum risk. All linear transformation weights are initialized using He et al. (2016)'s method, with the dropout rate set to $p = 0.4$.

Appendix B. Data sources and variable construction

Table B1

Detailed description, pre-processing method, and data sources for all variables.

Component	Variable	Description & Pre-processing	Unit	Data Source
Climate Hazards	Heat wave days	Annual days with a maximum temperature of $\geq 33^\circ\text{C}$. Raw 1 km grid data are downscaled to 500 m using area-weighted interpolation.	days	KMA extreme climate indices
	Tropical nights	Annual nights with a minimum temperature of $\geq 25^\circ\text{C}$. Raw 1 km grid data are downscaled to 500 m using area-weighted interpolation.	nights	KMA extreme climate indices
	Warm spell duration	Consecutive days where the daily mean temperature exceeds the 90th percentile for the reference period (1991–2020)	days	KMA extreme climate indices

(continued on next page)

Table B1 (continued)

Component	Variable	Description & Pre-processing	Unit	Data Source
Urban Exposure	Land surface temperature	Average 8-day summer (June–August) land surface temperature. Derived from MODIS Terra/Aqua Land Surface Temperature/Emissivity 8-Day L3 Global 1 km SIN Grid (MOD11A2 v006).	°C	NASA LP DAAC (MOD13A1.061)
	Population density	Residential population count per 500 m grid cell. Sourced directly from census grid data.	persons/km ²	Statistics Korea (SGIS)
	Impervious surface ratio	Proportion of impervious surfaces (urban, roads, etc.) calculated from a 30 m resolution land cover map, aggregated to a 500 m grid using an area-weighted average	%	EGIS Korea
	Building density (FAR)	Floor–area ratio, calculated as the total floor area divided by the total lot area for all buildings within a 500 m grid cell	%	National Spatial Data Platform
Social Sensitivity	Transport accessibility	Accessibility index normalized to a scale of 0–1, calculated by summing bus stops and subway entrances within a 500 m radius	dimensionless	Seoul Open Data Plaza
	Underground housing ratio	Ratio of semi-basement and basement housing units per 500 m grid. Calculated by geocoding official housing addresses for all years and aggregating counts per grid.	%	Address-based spatial analysis
	Young children	Percentage of the population aged 0–4 years within each 500 m grid.	%	National Spatial Data Platform
Adaptive Capacity	Low income	Percentage of Basic Livelihood Security Program recipients. Recipient counts at the administrative district (EMD) level are redistributed to 500 m grids using an area-weighted dasymetric method based on population grids.	%	Bokjiro OpenAPI
	Elderly population	Percentage of the population aged ≥65 years within each 500 m grid	%	National Spatial Data Platform
	Summer cooling electricity	Average summer (June–August) residential electricity consumption per household. District-level KEPCO data are joined with SGIS grid-level household counts.	kWh/household	KEPCO, SGIS
	Healthcare accessibility	Accessibility index normalized to a scale of 0–1, calculated using inverse distance weighting (1/distance) to the nearest general hospital	dimensionless	National Spatial Data Platform
	green space	Total area of parks within a 500 m walking distance divided by the residential population of the 500 m grid	m ² /person	National Spatial Data Platform
	Heatwave shelter density	A dimensionless index representing the number of public cooling facilities per 10,000 people within each 500 m grid. Sourced via the official National Disaster Safety Data Portal API.	dimensionless	National Disaster Safety Data Portal

Appendix C. Supplementary results

This appendix provides comprehensive descriptive statistics of the normalized risk scores ([0, 1] range) generated by the GAT model for Seoul's 2634 grids (resolution of 500 m) for 2015–2019 ($N = 13,170$ observations).

Table C1

Temporal risk statistics (2015–2019).

Year	Mean	SD	Median	Q1	Q3	Min.	Max.	N
2015	0.4292	0.1285	0.4274	0.312	0.652	0.083	0.982	2634
2016	0.4521	0.1249	0.4506	0.335	0.568	0.091	0.975	2634
2017	0.4184	0.1301	0.4170	0.298	0.538	0.088	0.968	2634
2018	0.4833	0.1215	0.4799	0.382	0.585	0.102	0.989	2634
2019	0.4208	0.1319	0.4186	0.305	0.541	0.086	0.978	2634
5-year average	0.4408	0.1294	0.4387	0.326	0.577	0.090	0.978	13,170

Note: The 5-year average shows a mean risk of 0.441 with moderate variability ($SD = 0.129$), indicating general mid-range vulnerability with significant spatial and temporal heterogeneity. The highest mean risk (0.483) was in 2018 due to record-breaking heat wave events in Seoul that summer.

Table C2

Regional risk statistics.

Region	Mean	SD	Median	Min.	Max.	Range (Max.–Min.)	N grids
Central Seoul	0.4676	0.1242	0.4679	0.195	0.752	0.557	1385
Northwest Seoul	0.4517	0.1289	0.4510	0.218	0.721	0.503	1515
Northeast Seoul	0.4472	0.1306	0.4460	0.182	0.708	0.526	3665
Southeast Seoul	0.4291	0.1297	0.4258	0.206	0.691	0.485	3050
Southwest Seoul	0.4290	0.1289	0.4298	0.193	0.698	0.505	3555

Note: N grids represents total grid-year observations for the 2015–2019 period (e.g., Central Seoul: 277 grids/year $\times$ 5 years = 1385 observations). Central Seoul exhibits the highest mean risk (0.468), driven by climate hazards, high urban exposure, and elevated social sensitivity. Southwest Seoul

has the lowest mean risk (0.429), reflecting lower urbanization and better green space access. The substantial intra-regional range (0.485–0.557) demonstrates significant within-district heterogeneity, validating the need for 500 m grid-level analysis.

To assess whether the scale-dependent spatial autocorrelation pattern is sensitive to the choice of graph connectivity parameter k , we computed Moran's I for the risk score and all four AR6 components under $k = 6, 8, 10$, and 12 using row-standardized k -nearest neighbor spatial weight matrices. All analyses used the 2015–2019 spatially aggregated mean values across 2634 grid cells. Coordinates were projected to the Korean national projection (EPSG:5179) prior to distance computation.

Table C3

Sensitivity of spatial autocorrelation (Moran's I) to graph connectivity parameter k .

k	Risk score	Hazard	Exposure	Sensitivity	Adaptive Capacity
6	0.507	0.977	0.759	0.869	0.572
8	0.498	0.972	0.738	0.853	0.563
10	0.486	0.967	0.710	0.835	0.549
12	0.478	0.962	0.688	0.817	0.534

Note: All values are statistically significant ($p < 0.001$, permutation test). The Hazard component consistently shows the highest Moran's I across all k values (0.962–0.977), confirming that climate hazard exhibits the strongest spatial concentration in Seoul regardless of graph topology. The dominance hierarchy (Hazard > Sensitivity > Exposure > Adaptive) and the absolute gap between Hazard and Exposure remain invariant across all tested k values, demonstrating that the scale-dependent transition in dominant risk factors reflects the intrinsic spatial structure of Seoul's urban climate rather than an artifact of the chosen graph construction parameter. The monotonic decrease in absolute Moran's I values with increasing k is an expected consequence of spatial smoothing over broader neighborhoods and does not affect the substantive finding of hierarchy invariance.

Table C4

Inter-method rank comparison across all 16 variables.

Variable	Component	SHAP	GNN	IG	GNN – SHAP	IG – SHAP
trans	Exposure	1	4	1	3	0
wdsi	Hazard	2	5	2	3	0
hw33	Hazard	3	12	5	9	2
baby	Sensitivity	4	3	4	1	0
inwater	Exposure	5	7	12	2	7
tr25	Hazard	6	1	3	5	3
poor	Sensitivity	7	2	13	5	6
popularity	Exposure	8	11	7	3	1
unlive	Sensitivity	9	10	11	1	2
area_ratio	Exposure	10	13	9	3	1
old	Sensitivity	11	9	8	2	3
lst	Hazard	12	6	14	6	2
cooler	Adaptive	13	15	10	2	3
green	Adaptive	14	14	15	0	1
cool	Adaptive	15	8	6	7	9
medical	Adaptive	16	16	16	0	0

Note: Ranks 1–16, where 1 = highest importance. |GNN – SHAP| and |IG – SHAP| denote absolute rank differences relative to SHAP. Mean |GNN – SHAP| by component: Climate Hazards = 5.75, Urban Exposure = 2.75, Social Sensitivity = 2.25, Adaptive Capacity = 2.25. Mean |IG – SHAP| by component: Adaptive Capacity = 3.25, Social Sensitivity = 2.75, Urban Exposure = 2.25, Climate Hazards = 1.75.

Data availability

The climate and spatial datasets used in this study are publicly available from the governmental and satellite data sources detailed in Appendix B (Table B1). Model-predicted heat risk scores, their descriptive statistics, and supplementary analyses are provided in Appendix C (Tables C1–C4). Health outcome data from the National Health Insurance Service (NHIS) and the Korea Disease Control and Prevention Agency (KDCA) are subject to privacy restrictions and are not publicly available. Researchers seeking access to health data must apply directly to the respective agencies. Model predictions and analysis results are available from the corresponding author upon reasonable request.

References

- Adadi, A., Berrada, M., 2018. Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). *IEEE Access* 6, 52138–52160. <https://doi.org/10.1109/ACCESS.2018.2870052>.
- Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: a next-generation hyperparameter optimization framework. In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 2623–2631. <https://doi.org/10.1145/3292500.3330701>.
- Anderson, B.G., Bell, M.L., 2009. Weather-related mortality: how heat, cold, and heat waves affect mortality in the United States. *Epidemiology* 20 (2), 205–213. <https://doi.org/10.1097/EDE.0b013e318190ee08>.

- Arrieta, A.B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., et al., 2020. Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf. Fusion* 58, 82–115. <https://doi.org/10.1016/j.inffus.2019.12.012>.
- Ba, J.L., Kiro, J.R., Hinton, G.E., 2016. Layer normalization. *arXiv*. <https://doi.org/10.48550/arXiv.1607.06450>, 1607.06450.
- Bae, Y., Joo, Y.M., 2020. The making of Gangnam: social construction and identity of urban place in South Korea. *Urban Aff. Rev.* 56 (3), 726–757. <https://doi.org/10.1177/1078087419827645>.
- Bai, J., Zhu, J., Song, Y., Zhao, L., Hou, Z., Du, R., Li, H., 2021. A3T-GCN: attention temporal graph convolutional network for traffic forecasting. *ISPRS Int. J. Geo Inf.* 10 (7), 485. <https://doi.org/10.3390/ijgi10070485>.
- Bao, J., Wang, Z., Yu, C., Li, X., 2016. The influence of temperature on mortality and its lag effect: a study in four Chinese cities with different latitudes. *BMC Public Health* 16 (1), 375. <https://doi.org/10.1186/s12889-016-3031-z>.
- Bengio, Y., Louradour, J., Collobert, R., Weston, J., 2009. Curriculum learning. In: *Proceedings of the 26th Annual International Conference on Machine Learning*, pp. 41–48. <https://doi.org/10.1145/1553374.1553380>.
- Bergmeir, C., Benítez, J.M., 2012. On the use of cross-validation for time series predictor evaluation. *Inf. Sci.* 191, 192–213. <https://doi.org/10.1016/j.ins.2011.12.028>.
- Bergstra, J., Bengio, Y., 2012. Random search for hyper-parameter optimization. *J. Mach. Learn. Res.* 13 (1), 281–305. <https://doi.org/10.5555/2188385.2188395>.
- Bhatt, U., Xiang, A., Sharma, S., Weller, A., Taly, A., Jia, Y., et al., 2020. Explainable machine learning in deployment. In: *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, pp. 648–657. <https://doi.org/10.1145/3351095.3375624>.
- Bowler, D.E., Buyung-Ali, L., Knight, T.M., Pullin, A.S., 2010. Urban greening to cool towns and cities: a systematic review of the empirical evidence. *Landsc. Urban Plan.* 97 (3), 147–155. <https://doi.org/10.1016/j.landurbplan.2010.05.006>.
- Burkart, K.G., Brauer, M., Aravkin, A.Y., Godwin, W.W., Hay, S.L., He, J., et al., 2021. Estimating the cause-specific relative risks of non-optimal temperature on daily mortality: a two-part modelling approach applied to the global burden of disease study. *Lancet* 398 (10301), 685–697. [https://doi.org/10.1016/S0140-6736\(21\)01700-1](https://doi.org/10.1016/S0140-6736(21)01700-1).
- Carvalho, C., Del Campo, A.G., de Carvalho Cabral, D., 2022. Scales of inequality: the role of spatial extent in environmental justice analysis. *Landsc. Urban Plan.* 221, 104369. <https://doi.org/10.1016/j.landurbplan.2022.104369>.
- Cawley, G.C., Talbot, N.L., 2010. On over-fitting in model selection and subsequent selection bias in performance evaluation. *J. Mach. Learn. Res.* 11, 2079–2107. <https://doi.org/10.5555/1756006.1859921>.
- Chakraborty, T., Hsu, A., Manya, D., Sheriff, G., 2019. Disproportionately higher exposure to urban heat in lower-income neighborhoods: a multi-city perspective. *Environ. Res. Lett.* 14 (10), 105003. <https://doi.org/10.1088/1748-9326/ab3b99>.
- Chen, Z., Badrinarayanan, V., Lee, C.Y., Rabinovich, A., 2018. GradNorm: gradient normalization for adaptive loss balancing in deep multitask networks. In: *Proceedings of the 35th International Conference on Machine Learning*, pp. 794–803. <https://doi.org/10.48550/arXiv.1711.02257>.
- Chen, T., Kornblith, S., Norouzi, M., Hinton, G., 2020. A simple framework for contrastive learning of visual representations. In: *Proceedings of the 37th International Conference on Machine Learning*, pp. 1597–1607. <https://doi.org/10.5555/3524938.3525087>.
- Cheng, W., Li, D., Liu, Z., Brown, R.D., 2021. Approaches for identifying heat-vulnerable populations and locations: a systematic review. *Sci. Total Environ.* 799, 149417. <https://doi.org/10.1016/j.scitotenv.2021.149417>.
- Chung, J., Gulcehre, C., Cho, K., Bengio, Y., 2014. Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv*. <https://doi.org/10.48550/arXiv.1412.3555>, 1412.3555.
- Confalonieri, R., Coba, L., Wagner, B., Besold, T.R., 2021. A historical perspective of explainable artificial intelligence. *Wiley Interdiscip. Rev. Data Min. Knowl. Discov.* 11 (1), e1391. <https://doi.org/10.1002/widm.1391>.
- Conlon, K.C., Mallen, E., Gronlund, C.J., Berrocal, V.J., Larsen, L., O'Neill, M.S., 2020. Mapping human vulnerability to extreme heat: a critical assessment of heat vulnerability indices created using principal components analysis. *Environ. Health Perspect.* 128 (9), 097001. <https://doi.org/10.1289/EHP4030>.
- D'Amour, A., Heller, K., Moldovan, D., Adlam, B., Alipanahi, B., Beutel, A., et al., 2022. Underspecification presents challenges for credibility in modern machine learning. *J. Mach. Learn. Res.* 23 (226), 1–61. <https://doi.org/10.5555/3586589.3586815>.
- Dong, W., Liu, Z., Zhang, L., Tang, Q., Liao, H., Li, X.E., 2014. Assessing heat health risk for sustainability in Beijing's urban heat island. *Sustainability* 6 (10), 7334–7357. <https://doi.org/10.3390/su6107334>.
- Doshi-Velez, F., Kim, B., 2017. Towards a rigorous science of interpretable machine learning. *arXiv*. <https://doi.org/10.48550/arXiv.1702.08608>, 1702.08608.
- Efron, B., Tibshirani, R.J., 1993. *An Introduction to the Bootstrap*. Chapman and Hall, New York.
- Fotheringham, A.S., Wong, D.W., 1991. The modifiable areal unit problem in multivariate statistical analysis. *Environ. Plan. A* 23 (7), 1025–1044. <https://doi.org/10.1068/a231025>.
- Gagne II, D.J., Haupt, S.E., Nychka, D.W., Thompson, G., 2019. Interpretable deep learning for spatial analysis of severe hailstorms. *Mon. Weather Rev.* 147 (8), 2827–2845. <https://doi.org/10.1175/MWR-D-18-0316.1>.
- Gal, Y., Ghahramani, Z., 2016. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In: *Proceedings of the 33rd International Conference on Machine Learning*, pp. 1050–1059. <https://doi.org/10.5555/3045390.3045502>.
- Hamilton, W.L., Ying, R., Leskovec, J., 2017. Representation learning on graphs: methods and applications. *arXiv*. <https://doi.org/10.48550/arXiv.1709.05584>, 1709.05584.
- Han, S., 2022. Spatial stratification and socio-spatial inequalities: the case of Seoul and Busan in South Korea. *Humanit. Soc. Sci. Commun.* 9 (1), 1–14. <https://doi.org/10.1057/s41599-022-01035-5>.
- Harvey, A.C., 1990. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, Cambridge, UK.
- He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778. <https://doi.org/10.1109/CVPR.2016.90>.
- Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., Meger, D., 2018. Deep reinforcement learning that matters. *Proc. AAAI Conf. Artif. Intell.* 32 (1), 3207–3214. <https://doi.org/10.1609/aaai.v32i1.11694>.
- Hinton, G., Vinyals, O., Dean, J., 2015. Distilling the knowledge in a neural network. *arXiv*. <https://doi.org/10.48550/arXiv.1503.02531>, 1503.02531.
- Hong, J.W., Hong, J., Kwon, E.E., Yoon, D.K., 2019. Temporal dynamics of urban heat island correlated with the socio-economic development over the past half-century in Seoul, Korea. *Environ. Pollut.* 254, 112934. <https://doi.org/10.1016/j.envpol.2019.07.102>.
- Huang, C.S., Lubell, M., Vantaggiato, F.P., 2023. Geographic scale dependency and the structure of climate adaptation policy networks in San Francisco Bay. *Ecol. Soc.* 28 (4). <https://doi.org/10.5751/ES-14403-280430>.
- Hyndman, R.J., Athanasopoulos, G., 2018. *Forecasting: Principles and Practice*, 2nd ed. OTexts, Melbourne, Australia. Available at <https://otexts.com/fpp2/>.
- IPCC, 2022. *Climate change 2022: impacts, adaptation and vulnerability*. In: *Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, UK.
- Jin, J., Kim, D., 2018. Expansion of the subway network and spatial distribution of population and employment in the Seoul metropolitan area. *Urban Stud.* 55 (11), 2499–2521. <https://doi.org/10.1177/0042098017725476>.
- Jin, G., Liang, Y., Fang, Y., Shao, Z., Huang, J., Zhang, J., Zheng, Y., 2023. Spatio-temporal graph neural networks for predictive learning in urban computing: a survey. *IEEE Trans. Knowl. Data Eng.* 36 (10), 5388–5408. <https://doi.org/10.1109/TKDE.2023.3333824>.
- Kang, J., Lyles, W., 2025. Understanding climate vulnerability and efficacy of cooling centers in urban “slum” housing neighborhoods: a case study of Jjokbang-chon, Seoul, South Korea. *Int. J. Disast. Risk Reduct.* 121, 105407. <https://doi.org/10.1016/j.ijdrr.2025.105407>.
- Karpatne, A., Atluri, G., Faghmous, J.H., Steinbach, M., Banerjee, A., Ganguly, A., et al., 2017. Theory-guided data science: a new paradigm for scientific discovery from data. *IEEE Trans. Knowl. Data Eng.* 29 (10), 2318–2331. <https://doi.org/10.1109/TKDE.2017.2720168>.
- Kendall, A., Gal, Y., Cipolla, R., 2018. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7482–7491. <https://doi.org/10.48550/arXiv.1705.07115>.

- Kenny, G.P., Yardley, J., Brown, C., Sigal, R.J., Jay, O., 2010. Heat stress in older individuals and patients with common chronic diseases. *CMAJ* 182 (10), 1053–1060. <https://doi.org/10.1503/cmaj.081050>.
- Keskitalo, C., Juhola, S., Westerhoff, L., 2016. Connecting multiple levels of governance for adaptation to climate change in advanced industrial states. In: Beunen, R., van Assche, K., Duineveld, M. (Eds.), *Evolutionary Governance Theory: An Introduction*. Springer, Cham, pp. 69–88. <https://doi.org/10.4324/9781315547626>.
- Kim, Y.H., Baik, J.J., 2005. Spatial and temporal structure of the urban heat island in Seoul. *J. Appl. Meteorol.* 44 (5), 591–605. <https://doi.org/10.1175/JAM2226.1>.
- Klemmer, K., Safir, N.S., Neill, D.B., 2023. Positional encoder graph neural networks for geographic data. In: *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, pp. 1379–1389. <https://doi.org/10.48550/arXiv.2111.10144>.
- Kruskal, W.H., Wallis, W.A., 1952. Use of ranks in one-criterion variance analysis. *J. Am. Stat. Assoc.* 47 (260), 583–621.
- Lara, M.J., McGuire, A.D., Euskirchen, E.S., Genet, H., Yi, S., Rutter, R., et al., 2020. Local-scale Arctic tundra heterogeneity affects regional-scale carbon dynamics. *Nat. Commun.* 11 (1), 4925. <https://doi.org/10.1038/s41467-020-18768-z>.
- Le Rest, K., Pinaud, D., Monestiez, P., Chadoeuf, J., Bretagnolle, V., 2014. Spatial leave-one-out cross-validation for variable selection in the presence of spatial autocorrelation. *Glob. Ecol. Biogeogr.* 23 (7), 811–820. <https://doi.org/10.1111/geb.12161>.
- Li, F., Yigitcanlar, T., Nepal, M., Thanh, K.N., Dur, F., 2024. A novel urban heat vulnerability analysis: integrating machine learning and remote sensing for enhanced insights. *Remote Sens.* 16 (16), 3032. <https://doi.org/10.3390/rs16163032>.
- Liu, J., Kim, H., Hashizume, M., Lee, W., Honda, Y., Kim, S.E., et al., 2025. Nonlinear exposure-response associations of daytime, nighttime, and day-night compound heatwaves with mortality amid climate change. *Nat. Commun.* 16 (1), 635. <https://doi.org/10.1038/s41467-025-56067-7>.
- Lundberg, S.M., Lee, S.L., 2017. A unified approach to interpreting model predictions. *Adv. Neural Inf. Proces. Syst.* 30, 4765–4774. <https://doi.org/10.48550/arXiv.1705.07874>.
- Martin, R., Sunley, P., 2006. Path dependence and regional economic evolution. *J. Econ. Geogr.* 6 (4), 395–437. <https://doi.org/10.1093/jeg/ibl012>.
- Melis, G., Di Gangi, E., Ellena, M., Zengarini, N., Ricciardi, G., Mercogliano, P., Costa, G., 2023. Urban Heat Island effect and social vulnerability in Turin: prioritizing climate change mitigation action with an equity perspective. *Sci. Talks* 8, 100258. <https://doi.org/10.1016/j.sctalk.2023.100258>.
- Miller, T., 2019. Explanation in artificial intelligence: insights from the social sciences. *Artif. Intell.* 267, 1–38. <https://doi.org/10.1016/j.artint.2018.07.007>.
- Miyato, T., Kataoka, T., Koyama, M., Yoshida, Y., 2018. Spectral normalization for generative adversarial networks. *arXiv*. <https://doi.org/10.48550/arXiv.1802.05957>, 1802.05957.
- Ngarame, J., Nganyiyimana, J., Kim, I., Santamouris, M., Yun, G.Y., 2020. Synergies between urban heat island and heat waves in Seoul: the role of wind speed and land use characteristics. *PLoS One* 15 (12), e0243571. <https://doi.org/10.1371/journal.pone.0243571>.
- Niu, Y., Li, Z., Gao, Y., Liu, X., Xu, L., Vardoulakis, S., Liu, Q., 2021. A systematic review of the development and validation of the heat vulnerability index: major factors, methods, and spatial units. *Curr. Clim. Chang. Rep.* 7 (3), 87–97. <https://doi.org/10.1007/s40641-021-00173-3>.
- Ou, Y., Kim, E., Liu, X., Nam, K.M., 2023. Delineating functional regions from road networks: the case of South Korea. *Environ. Plan. B Urban Anal. City Sci.* 50 (6), 1677–1694. <https://doi.org/10.1177/23998083231172198>.
- Pascanu, R., Mikolov, T., Bengio, Y., 2013. On the difficulty of training recurrent neural networks. In: *Proceedings of the 30th International Conference on Machine Learning*, pp. 1310–1318. <https://doi.org/10.5555/3042817.3043083>.
- Ploton, P., Mortier, F., Réjou-Méchain, M., Barbier, N., Picard, N., Rossi, V., et al., 2020. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nat. Commun.* 11 (1), 4540. <https://doi.org/10.1038/s41467-020-18321-y>.
- Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* 378, 686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>.
- Reid, C.E., O'Neill, M.S., Gronlund, C.J., Brines, S.J., Brown, D.G., Diez-Roux, A.V., Schwartz, J., 2009. Mapping community determinants of heat vulnerability. *Environ. Health Perspect.* 117 (11), 1730–1736. <https://doi.org/10.1289/ehp.0900683>.
- Reimers, N., Gurevych, I., 2017. Reporting score distributions makes a difference: performance study of LSTM-networks for sequence tagging. *arXiv*. <https://doi.org/10.48550/arXiv.1707.09861>, 1707.09861.
- Ribeiro, M.T., Singh, S., Guestrin, C., 2016. “Why should I trust you?” Explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144. <https://doi.org/10.1145/2939672.2939778>.
- Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillera-Aroita, G., et al., 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40 (8), 913–929. <https://doi.org/10.1111/ecog.02881>.
- Rosenzweig, C., Solecki, W.D., Romero-Lankao, P., Mehrotra, S., Dhakal, S., Ibrahim, S.A. (Eds.), 2018. *Climate Change and Cities: Second Assessment Report of the Urban Climate Change Research Network*. Cambridge University Press, Cambridge, UK.
- Rudin, C., 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat. Mach. Intell.* 1 (5), 206–215. <https://doi.org/10.1038/s42256-019-0048-x>.
- Samek, W., Wiegand, T., Müller, K.R., 2017. Explainable artificial intelligence: understanding, visualizing and interpreting deep learning models. *arXiv*. <https://doi.org/10.48550/arXiv.1708.08296>, 1708.08296.
- Samek, W., Montavon, G., Lapuschkin, S., Anders, C.J., Müller, K.R., 2021. Explaining deep neural networks and beyond: a review of methods and applications. *Proc. IEEE* 109 (3), 247–278. <https://doi.org/10.1109/JPROC.2021.3060483>.
- Schlosberg, D., 2012. Climate justice and capabilities: a framework for adaptation policy. *Ethics Int. Aff.* 26 (4), 445–461. <https://doi.org/10.1017/S0892679412000615>.
- Sining, C., Jun, G., 2021. Climatic risks of Beijing–Tianjin–Hebei urban agglomeration and their changes. *Geomat. Nat. Hazards Risk* 12 (1), 1298–1314. <https://doi.org/10.1080/19475705.2021.1924296>.
- Snoek, J., Larochelle, H., Adams, R.P., 2012. Practical Bayesian optimization of machine learning algorithms. *Adv. Neural Inf. Proces. Syst.* 25, 2951–2959. <https://doi.org/10.48550/arXiv.1206.2944>.
- Son, J.Y., Lee, J.T., Anderson, G.B., Bell, M.L., 2012. The impact of heat waves on mortality in seven major cities in Korea. *Environ. Health Perspect.* 120 (4), 566–571. <https://doi.org/10.1289/ehp.1103759>.
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15 (1), 1929–1958. <https://doi.org/10.5555/2627435.2670313>.
- Stewart, I.D., Oke, T.R., 2012. Local climate zones for urban temperature studies. *Bull. Am. Meteorol. Soc.* 93 (12), 1879–1900. <https://doi.org/10.1175/BAMS-D-11-00019.1>.
- Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., et al., 2015. Going deeper with convolutions. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1–9. <https://doi.org/10.1109/CVPR.2015.7298594>.
- Szekely, G.J., Rizzo, M.L., Bakirov, N.K., 2007. Measuring and testing dependence by correlation of distances. *Ann. Stat.* 35 (6), 2769–2794. <https://doi.org/10.1214/009053607000000505>.
- Tashman, L.J., 2000. Out-of-sample tests of forecasting accuracy: an analysis and review. *Int. J. Forecast.* 16 (4), 437–450. [https://doi.org/10.1016/S0169-2070\(00\)00065-0](https://doi.org/10.1016/S0169-2070(00)00065-0).
- Tomlinson, C.J., Chapman, L., Thornes, J.E., Baker, C.J., 2011. Including the urban heat island in spatial heat health risk assessment strategies: a case study for Birmingham, UK. *Int. J. Health Geogr.* 10 (1), 42. <https://doi.org/10.1186/1476-072X-10-42>.
- Varma, S., Simon, R., 2006. Bias in error estimation when using cross-validation for model selection. *BMC Bioinform.* 7 (1), 91. <https://doi.org/10.1186/1471-2105-7-91>.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., et al., 2017. Attention is all you need. *Adv. Neural Inf. Proces. Syst.* 30, 5998–6008. <https://doi.org/10.48550/arXiv.1706.03762>.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., Bengio, Y., 2017. Graph attention networks. *arXiv*. <https://doi.org/10.48550/arXiv.1710.10903>, 1710.10903.

- Veličković, P., Fedus, W., Hamilton, W.L., Liò, P., Bengio, Y., Hjelm, R.D., 2018. Deep graph infomax. arXiv. <https://doi.org/10.48550/arXiv.1809.10341>, 1809.10341.
- Vicedo-Cabrera, A.M., Scovronick, N., Sera, F., Royé, D., Schneider, R., Tobias, A., et al., 2021. The burden of heat-related mortality attributable to recent human-induced climate change. *Nat. Clim. Chang.* 11 (6), 492–500. <https://doi.org/10.1038/s41558-021-01058-x>.
- Vincent, P., Larochelle, H., Bengio, Y., Manzagol, P.A., 2008. Extracting and composing robust features with denoising autoencoders. In: In: Proceedings of the 25th International Conference on Machine Learning, pp. 1096–1103. <https://doi.org/10.1145/1390156.1390294>.
- Wang, X., Chen, Y., Zhu, W., 2021. A survey on curriculum learning. *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (9), 4555–4576. <https://doi.org/10.1109/TPAMI.2021.3069908>.
- Willard, J., Jia, X., Xu, S., Steinbach, M., Kumar, V., 2022. Integrating scientific knowledge with machine learning for engineering and environmental systems. *ACM Comput. Surv.* 55 (4), 66. <https://doi.org/10.1145/3514228>.
- World Health Organization, 2024. Heat and Health. Fact Sheet. Available at: <https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health> (Accessed: 28 May 2024).
- Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., Yu, P.S., 2020. A comprehensive survey on graph neural networks. *IEEE Trans. Neural Netw. Learn. Syst.* 32 (1), 4–24. <https://doi.org/10.1109/TNNLS.2020.2978386>.
- Xiong, R., Yang, Y., He, D., Zheng, K., Zheng, S., Xing, C., et al., 2020. On layer normalization in the transformer architecture. In: Proceedings of the 37th International Conference on Machine Learning, PMLR, pp. 10524–10533. <https://doi.org/10.48550/arXiv.2002.04745>.
- Yi, C., Kwon, H.G., Yang, H., 2022. Spatial temperature differences in local climate zones of Seoul metropolitan area during a heatwave. *Urban Clim.* 41, 101012. <https://doi.org/10.1016/j.uclim.2021.101012>.
- Ying, Z., Bourgeois, D., You, J., Zitnik, M., Leskovec, J., 2019. GNNExplainer: generating explanations for graph neural networks. *Adv. Neural Inf. Proces. Syst.* 32, 9240–9251. <https://doi.org/10.48550/arXiv.1903.03894>.
- Yuan, H., Yu, H., Gui, S., Ji, S., 2022. Explainability in graph neural networks: a taxonomic survey. *IEEE Trans. Pattern Anal. Mach. Intell.* 45 (5), 5782–5799. <https://doi.org/10.1109/TPAMI.2022.3204236>.
- Zhao, L., Song, Y., Zhang, C., Liu, Y., Wang, P., Lin, T., et al., 2019. T-GCN: a temporal graph convolutional network for traffic prediction. *IEEE Trans. Intell. Transp. Syst.* 21 (9), 3848–3858. <https://doi.org/10.1109/TITS.2019.2935152>.